

Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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Introduction

Cluster randomized trials (CRTs) randomize intact groups, such as clinics, schools, or geographic areas, rather than individuals, and are the standard design when an intervention is delivered at the group level or when individual randomization would create unacceptable contamination between arms [1, 2]. A central assumption underlying standard CRT analysis is the no-interference component of the stable unit treatment value assumption (SUTVA): a cluster’s outcome must not depend on which other clusters were assigned to treatment [3]. This assumption, borrowed from individually randomized trial theory, can fail once clusters are spatially arranged, and so can the weaker assumption of partial interference, under which outcomes may depend on assignments within prespecified groups but not across them [4]. Tobler’s first law of geography, “everything is related to everything else, but near things are more related than distant things” [5], implies that an intervention delivered in one location can plausibly influence outcomes in geographically adjacent locations through shared populations, service networks, information diffusion, or environmental pathways. Documented examples of this kind of spatial spillover span public health and development economics: in a trial of insecticide-treated bed nets in northern Ghana, child mortality was lower in control villages close to treated villages than in

control villages farther away [6], and a school-based deworming trial in Kenya found health and attendance benefits among untreated children near treated schools [7]. When control clusters benefit from nearby treatment in this way, an unadjusted comparison of treated and control clusters may understate the direct effect, and how much contamination occurs can depend on the spatial arrangement of treated and control clusters, not only on the number of clusters allocated to each arm.

The design-stage gap in the spatial interference literature

The methodological literature addressing spatial interference in CRTs has concentrated on analysis-stage correction. Spatial autocorrelation statistics such as Moran’s I [8, 9] and Geary’s C [10] provide tools for detecting residual spatial dependence once outcomes have been observed, and spatial regression models, including spatial lag and spatial Durbin specifications [11], estimated by maximum likelihood [12] and implemented in modern software [13], allow an analyst to separate a direct treatment effect from a spillover effect after the fact. In their systematic review of spatial analysis methods for cluster randomized trials, Jarvis and colleagues placed trials that used spatial separation or buffers outside the primary review of statistical methods [14]. The literature therefore offers much more guidance on analyzing spatial effects than on how the choice of treatment assignment itself, made before any data are collected, might reduce the problem.

Design-stage tools do exist, but they were largely developed for a different purpose. Restricted randomization addresses the baseline imbalance that a limited number of clusters can produce [15]. Covariate-constrained randomization generates candidate assignments, retains those meeting a prespecified covariate-balance criterion, and randomizes among them [16]; Crisp and colleagues extended it to the selection of clusters from a larger eligible set and noted that their procedure also increases spatial dispersion among the selected clusters [17]. Their dispersion criterion concerns which clusters are selected rather than how the selected clusters are assigned to

arms, and none of these methods targets the spatial pattern of arm assignment under spillover: balancing a covariate across arms does not by itself determine whether treated and control clusters are spatially interspersed in a way that induces contamination. A more recent body of work addresses the problem studied in this chapter more directly. Randomized saturation designs assign different saturations, the fractions of units treated, to different groups and randomize treatment within each group, which allows direct and spillover effects to be jointly identified rather than merely controlled for [18]. A conventional “fried-egg” design omits each cluster’s outer area from analysis to limit contamination; McCann and colleagues proposed instead selecting whole clusters for inclusion or exclusion while preserving randomization [19]. Most recently, a group of theoretical papers published or posted between 2022 and 2025 has derived rate-optimal cluster-randomized designs under spatial interference [20], independent-set designs under network interference [21], designs that explicitly account for cross-cluster interference [22], and designs for spatio-temporally heterogeneous treatment effects [23]. This chapter complements this work with a simulation comparison of concrete assignment strategies that a trial team could implement, under heterogeneous baseline incidence and a range of spatial dependence and spillover conditions.

Extending a small-grid proof of concept

This chapter extends the proof-of-concept study of Chapter 2, which asked a narrower version of the same question: how does a priori treatment-assignment strategy affect estimation under spatial spillover and spatial dependence? That study compared simple random assignment against a single block-stratified design, on small, exhaustively enumerable grids of 8 to 12 spatial units arranged in 2×4 , 3×3 , and 3×4 configurations, and estimated the intervention effect with a spatial autoregressive (SAR) lag model containing a spillover indicator, with separate parameters for a baseline effect, the intervention effect, the spillover effect, and spatial dependence. Among its future considerations, Chapter 2 listed a broader range of grid dimensions and neighbor

structures beyond rook contiguity, and it proposed comparing simple random assignment and block-stratified assignment with buffer-zone (“fried-egg”) designs, which exclude from analysis the control clusters that border treated clusters. This chapter takes up a larger grid and queen contiguity, and it adds two extensions: several further assignment designs, none of which excludes clusters from analysis, and heterogeneous, rather than spatially uniform, baseline outcome risk. Chapter 2 suggested weighting assignment by cluster population density; some of the designs here use baseline outcome incidence instead. We scale from 8–12 clusters to a 100-cluster grid, from two designs to six conceptually distinct treatment assignment strategies (two further variants, which could not be distinguished from a retained design, are reported in Appendix A1), and from a single, uniform baseline risk surface to three distinct mechanisms of heterogeneous baseline incidence. This lets us ask which designs estimate the treatment effect well when baseline risk varies across space. Some of the designs use the incidence surface to assign treatment, and others ignore it.

Aims

This chapter has three aims. The first aim is to rank the six treatment assignment designs by estimation accuracy, measured by mean squared error (MSE), and by inferential validity, measured by 95% confidence interval coverage, separately within each of the three incidence-heterogeneity mechanisms and their five configurations, since a design that performs well under one mechanism may not generalize to another. The second aim is to test whether that ranking is stable across the spatial and spillover parameters that a real trial team cannot control or know in advance, including the magnitude of the true treatment effect itself, the strength of residual spatial dependence in the outcome, and the magnitude of the spillover effect. The third aim is to translate the resulting ranking into design guidance and describe how it will be applied in a planned sudden unexpected death (SUD) prevention pilot to be delivered through North Carolina’s community college network. The sections that follow describe the simulation methods,

report results for each aim, describe the planned SUD application, and discuss limitations and future work.

Methods

Spatial structure and outcome model

We simulated a 10×10 regular grid of 100 clusters under two definitions of adjacency. Under queen contiguity, two clusters are neighbors if they share an edge or a corner; under rook contiguity, they must share an edge. Queen contiguity is the primary analysis in this chapter. Rook contiguity is a sensitivity analysis and the case that continues Chapter 2, whose grids used rook neighbors. For each definition, the binary adjacency matrix was row-standardized to form the spatial weights matrix W , so that WZ gives, for each cluster, the proportion of its neighbors that are treated, and WY gives the average outcome among its neighbors. Row-standardization expresses each cluster's spillover exposure as an average over its own neighborhood rather than a raw sum, which would otherwise depend on how many neighbors the cluster has. Within a scenario, the same neighbor type defines W in the outcome model, in the Isolation Buffer design, and in the fitted model.

Cluster-level outcomes Y were generated from a spatial lag model with a spillover term,

$$Y = (I - \rho W)^{-1} [\tau Z + S(Z) + \beta X + \epsilon], \quad \epsilon \sim N(0, \sigma^2 I), \quad (1)$$

where $Z \in \{0, 1\}^{100}$ is the treatment assignment produced by the design under evaluation, τ is the direct treatment effect of interest, $S(Z)$ is the spillover term defined in Equation 2, ρ is the spatial autoregressive coefficient governing dependence in the outcome beyond that induced by treatment and spillover, X is the baseline incidence covariate with coefficient $\beta = 1$, and $\sigma = 1$. The spatial multiplier $(I - \rho W)^{-1}$ propagates each cluster's shocks through the network

121 of neighbors. We tested $\rho \in \{0, 0.01, 0.20, 0.50\}$. When $\rho = 0$ the multiplier is the identity and
 122 the model is a linear regression with independent errors, $Y = \tau Z + S(Z) + \beta X + \epsilon$, in which
 123 spatial structure enters only through the spillover term; larger values of ρ induce progressively
 124 stronger spatial clustering in the outcome.

125 We evaluated two spillover regimes,

$$S(Z) = \gamma WZ \quad (\text{both arms}), \quad S(Z) = \gamma WZ \circ (1 - Z) \quad (\text{control only}), \quad (2)$$

126 where $\circ$ denotes the elementwise product. Under the control-only regime, only untreated clusters
 127 respond to the treatment status of their neighbors, which represents classical contamination:
 128 an untreated cluster's outcome rises with the share of its neighbors that are treated. Under
 129 the both-arms regime, every cluster responds to its neighbors' treatment regardless of its
 130 own assignment, so neighboring treatment adds to a treated cluster's own effect. We tested
 131 four spillover magnitudes, $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, and five magnitudes of the direct effect,
 132 $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, with $\tau = 1$ as the primary scenario throughout.

133 **Heterogeneous baseline incidence**

134 Chapter 2 assumed a spatially uniform baseline outcome risk. Cluster-level health outcomes
 135 typically vary across space, and the designs studied here differ in whether and how they use that
 136 variation. We generated the baseline incidence covariate X , with values in $(0, 1)$, under three
 137 mechanisms. Under the independent (iid) mechanism, each X_i is drawn from a $\text{Uniform}(0, 1)$
 138 distribution, with no spatial structure. Under the spatially autocorrelated mechanism, $X =$
 139 $\Phi\{(I - \rho_X W_Q)^{-1}u\}$, where $u \sim N(0, I)$, Φ is the standard normal distribution function applied
 140 elementwise, W_Q is the queen weights matrix, and $\rho_X \in \{0.20, 0.50\}$. Under the Poisson
 141 mechanism, a spatially autocorrelated log-rate surface, $\log \lambda = \log(35/100,000) + (I - \rho_X W_Q)^{-1}u$
 142 with $\rho_X \in \{0.20, 0.50\}$, generates cluster death counts $C_i \sim \text{Poisson}(\lambda_i P)$, with an equal

143 population of $P = 100,000$ in every cluster, so that a cluster at the base rate expects 35 deaths.
 144 The observed rates C_i/P are then rank-normalized to $X_i = (r_i - 0.5)/N$, where r_i is the rank
 145 of cluster i 's rate among the $N = 100$ clusters and tied rates share their average rank. Ties
 146 remain at this population size: the Poisson surfaces used here have between 60 and 76 distinct
 147 values among the 100 clusters. The base rate of 35 per 100,000 was chosen to be representative
 148 of published SUD incidence: it lies between the SUD incidence rates that Mirzaei and colleagues
 149 estimated for US men (49.3 per 100,000) and women (21.7 per 100,000) aged 20–64, close to their
 150 average [24]. It is not the observed rate of the county data described in the Application section,
 151 which come from a different source, age range, period, and case definition. The incidence filters
 152 always use the queen weights W_Q , whichever neighbor type the scenario uses, so the incidence
 153 surfaces are identical under rook and queen contiguity. The iid mechanism and the two strengths
 154 of each of the other two mechanisms give five incidence configurations.

155 For each configuration we generated $K = 10$ incidence surfaces, $X_1, \dots, X_{10}$, each from its own
 156 seed. The same ten surfaces are used by every scenario in that configuration, across neighbor
 157 types, outcome-model parameters, designs, and values of τ ; the design draws and the outcome
 158 noise are shared more selectively, as described under Simulation design, resampling, and metrics.
 159 Each design is applied to the surface that generates the outcome: assignments for surface k
 160 are drawn from X_k , and the outcome and the fitted model for those assignments use X_k as
 161 well. This assumes that the incidence observed when a trial is planned is the incidence that
 162 drives its outcome (the matched-surface assumption). The target of each scenario is performance
 163 averaged over incidence surfaces; for example, $\text{MSE} = E_X E_{Z|X} E_\epsilon [(\hat{\tau} - \tau)^2]$, with X drawn from
 164 the configuration's generator, Z from the design given X , and $\epsilon \sim N(0, I)$. Bias, coverage, and
 165 power are defined in the same way.

Treatment assignment designs

We implemented eight treatment assignment designs and present six of them as the primary comparison set in this chapter. The other two, Saturation Quadrants and Balanced Halves, each share the mechanism of a retained design, and in the eight-design comparison at $\tau = 1$ neither could be distinguished from its counterpart under either neighbor type, by the Nemenyi and Holm-adjusted Wilcoxon signed-rank tests described under Simulation design, resampling, and metrics below, applied here to all eight designs. Saturation Quadrants assigns the saturations 20%, 40%, 60%, and 80% to the four quadrants in random order, without reference to incidence. Its mean squared error was 0.0900, against 0.0907 for Incidence-Guided Saturation Quadrants, under queen contiguity (Nemenyi $p = 0.98$; Holm-adjusted Wilcoxon signed-rank $p = 0.41$), and 0.0708 against 0.0717 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). We retain one representative of the saturation mechanism, Incidence-Guided Saturation Quadrants. Saturation Quadrants had the marginally lower mean under both neighbor types and the better average rank in the eight-design comparison (2.100 against 2.350 under queen contiguity and 2.281 against 2.306 under rook contiguity), so this choice does not rest on performance; that ordering the saturations by incidence makes no detectable difference is itself a result of this chapter. Balanced Halves divides clusters into two halves by incidence rank and treats 25 at random in each. Its mean squared error was 0.1299, against 0.1308 for Balanced Quartiles, under queen contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.88$), and 0.0842 against 0.0837 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). The two designs differ only in how finely they stratify on incidence, and we retain the finer, four-stratum version. Balanced Halves had the marginally lower mean and the better average rank under queen contiguity (3.312 against 3.356), and Balanced Quartiles had both under rook contiguity (2.919 against 3.038), so this choice does not rest on performance either. Both excluded designs, and the full eight-design comparison, are reported in Appendix A1.

Table 1 describes the six retained designs, and Figure 1 shows one assignment from each on a single illustrative incidence surface. Checkerboard is the strict block-stratified allocation of Chapter 2, in which no two rook neighbors share an arm. Chapter 2’s Block Stratified Sampling design also admitted relaxed allocations on its 3×3 grid, where the arms cannot be balanced; on its 2×4 and 3×4 grids, as on the 10×10 grid here, the strict allocations are the two checkerboard patterns. We use the two names interchangeably for this allocation and use Checkerboard in the rest of this chapter. Checkerboard is the only deterministic design: it gives the same assignment in every draw, although each of its fits still receives its own outcome noise. Every other design was drawn independently 25 times for each incidence surface. Designs that rank clusters by incidence break ties at random, afresh for each draw. High Incidence Focus is random only through that tie-breaking; under the iid and spatial mechanisms incidence values are distinct, so it gives the same assignment in all 25 draws on a surface, and its assignment changes only from one surface to the next. Isolation Buffer builds its assignment from the neighbor type of the scenario, and it treats 38 clusters on average under rook contiguity and 21 under queen contiguity. Every other design treats exactly 50.

Estimation

We estimated τ with two maximum likelihood spatial lag models, both fit to every simulated outcome and both including an intercept. The primary, oracle model has the spatial lag ρWY and the covariates Z , $\text{Spill} = S(Z)$, and X_k , so it matches the outcome model in Equation 1, including the true regime-specific spillover covariate from Equation 2. The oracle model assumes that the analyst knows the form of the spillover mechanism and has the exact exposure term, so its results are a best case for each design; we return to this assumption as a limitation. The non-oracle model, a sensitivity analysis, has the covariates Z and X_k only and omits the spillover term. Both models are fit to the same outcomes.

We fit both models with our own implementation of the maximum likelihood estimator of

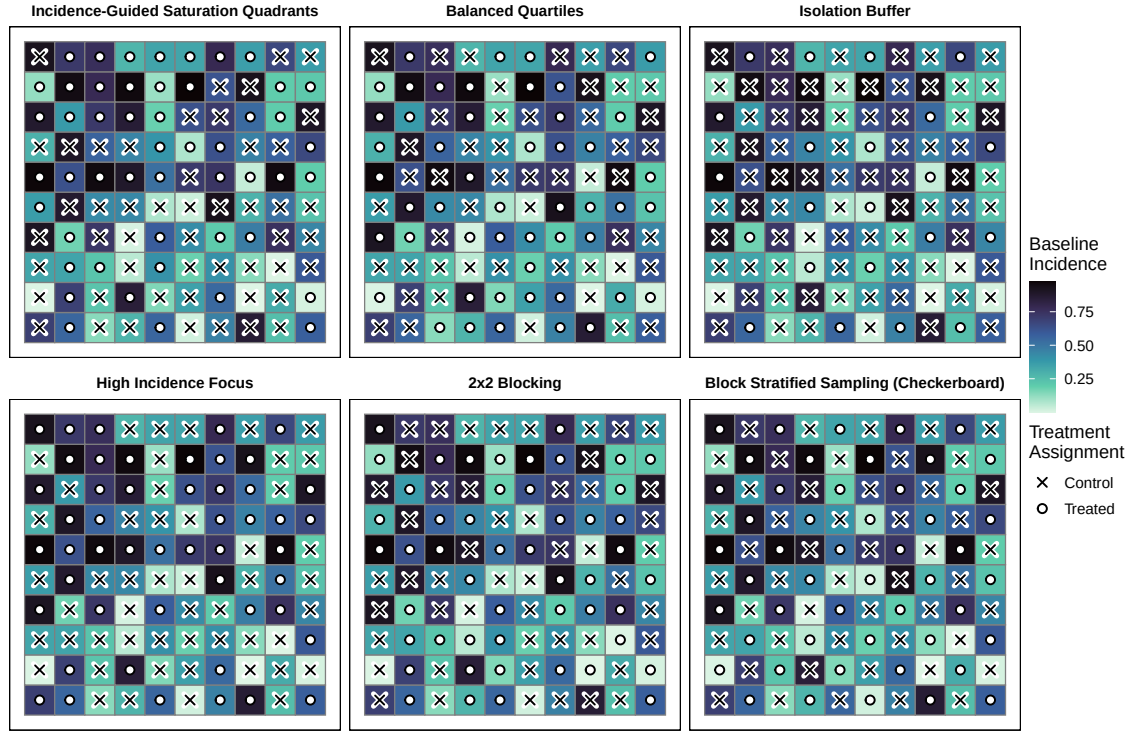


Figure 1: One sample assignment from each of the six designs on the 10×10 grid, with panels in the order of Table 1. All six assignments use a single illustrative incidence surface of 100 independent $\text{Uniform}(0, 1)$ values generated with a fixed seed; it is not one of the simulation's incidence surfaces. Tile fill shows baseline incidence (darker is higher), a ring marks a treated cluster, and an X marks a control. Isolation Buffer is shown under queen contiguity. The Checkerboard panel is labeled Block Stratified Sampling (Checkerboard), the Chapter 2 name for this allocation (see Treatment assignment designs).

Table 1: The six treatment assignment designs compared in this chapter, ordered by mean squared error under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (Table 2). A cluster’s incidence rank is its rank by the baseline incidence surface X_k , with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment; for Isolation Buffer, which depends on the neighbor type, it is the mean over its assignments under each neighbor type.

Design	Assignment rule	Clusters treated	Deterministic
Incidence-Guided Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants. Quadrants are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Quartiles	Clusters are divided by incidence rank into four strata of 25. Twelve clusters are treated at random in each stratum, and one more in each of two randomly chosen strata	50	No
Isolation Buffer	Clusters are treated one at a time, each chosen at random from those that are neither treated nor adjacent to a treated cluster, until none remain (a greedy random maximal independent set); no two treated clusters are neighbors	38 rook, 21 queen (mean)	No
High Incidence Focus	The 50 clusters with the highest incidence rank are treated	50	No
2×2 Blocking	The grid is divided into 25 non-overlapping 2×2 blocks; two of the four clusters in each block are treated at random	50	No
Checkerboard (strict Block Stratified Sampling allocation in Chapter 2)	Clusters whose grid coordinates (x, y) have an odd sum $x + y$ are treated, so treatment alternates from cluster to cluster; under rook contiguity every neighbor of a treated cluster is a control, and every neighbor of a control is treated	50	Yes

the spatial lag model [25], the same estimator that the `lagsarlm` function of the R package `spatialreg` computes with `method = "eigen"`. It maximizes the concentrated log-likelihood over ρ , computing the log-determinant from the eigenvalues of W [12], estimates the remaining coefficients by least squares given $\hat{\rho}$, and computes standard errors from the analytic information matrix. On 5,120 validation fits it agreed with `lagsarlm` to within 9.7×10^{-8} in $\hat{\tau}$ and to a relative difference of 1.1×10^{-8} in its standard error, at roughly 1% of the computing time; we also refit a 1% sample of the full simulation with `lagsarlm` itself as a check. Confidence intervals are 95% Wald intervals, $\hat{\tau} \pm 1.96 \widehat{SE}$.

Two combinations of design and neighbor type weaken or remove the information that separates

τ from the spillover term. Under rook contiguity, Checkerboard gives $WZ = 1 - Z$ exactly, because every neighbor of a treated cluster is a control and every neighbor of a control is treated. Both spillover covariates then equal $\gamma(1 - Z)$, a linear combination of the intercept and Z , so τ is not identified: the fitting routine drops the aliased spillover column, and the coefficient on Z estimates $\tau - \gamma$. Chapter 2 noted the same consequence of exact collinearity between treatment and spillover exposure, $\hat{\beta} \approx \beta - \psi$ in its notation. The non-oracle model has no spillover column to drop, but its Checkerboard estimates under rook contiguity have the same defect. Under queen contiguity, each of the 64 interior clusters has four treated and four control neighbors, so $WZ = 1/2$ at every interior cluster. Under the control-only regime the spillover covariate then equals $(\gamma/2)(1 - Z)$ at those clusters, a linear combination of the intercept and Z , and only the 36 clusters on the grid boundary carry information that separates it from the intercept and Z ; τ is identified, but imprecisely. Under the both-arms regime it is the constant $\gamma/2$ at interior clusters, confounded with the intercept but not with Z . Isolation Buffer under rook contiguity can also, rarely, draw exactly the checkerboard pattern, a maximal independent set that equals one color class of the grid. This happened in 4 of its 40,000 rook draws and affected one fit in each of 20 scenarios, 4 at each value of τ . We flag every fit in which the fitting routine reports an aliased column and every scenario in which some draw has $\text{rank}([1, Z, WZ]) < 3$. Aliased estimates are kept in all summaries. In the exhibits, we flag Checkerboard under rook contiguity, where every fit is aliased; the Isolation Buffer scenarios, each with one aliased fit among its 250, are noted here and carry no flag in the exhibits.

Simulation design, resampling, and metrics

The full simulation crossed five values of τ , five incidence configurations, two neighbor types, four values of ρ , four values of γ , two spillover regimes, and eight designs, for $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12,800$ scenarios, of which the six designs reported in this chapter account for 9,600. We call a combination of incidence configuration, neighbor type, ρ , γ , and spillover regime a block; there

are $5 \times 2 \times 4 \times 4 \times 2 = 320$ blocks, 160 for each neighbor type. At the primary scenario, $\tau = 1$, the six designs give 1,920 scenarios, and every block contains all six.

Within each scenario, we drew $J = 25$ assignments from each of the $K = 10$ incidence surfaces, for 250 fits per scenario, and gave every fit its own noise vector, so that fit (k, j) has outcome $Y_{kj} = (I - \rho W)^{-1}[\tau Z_{kj} + S(Z_{kj}) + \beta X_k + \epsilon_{kj}]$. Checkerboard’s 250 fits share one assignment but have 250 distinct noise vectors. Given a surface, its 25 pairs of assignment and noise are independent, and the ten surfaces are independent of one another, so the surface is the primary sampling unit. Each estimator was fit 3.2 million times.

Every random draw was seeded from a hash of a key that names the draw: the configuration and surface index for an incidence surface; the block, surface, and design for a set of 25 assignments; and the block for the noise. Each draw therefore depends only on its key, and the results are identical whether scenarios are run sequentially, in parallel, or in any order. The keys also give common random numbers. Within a block, all designs share the same noise vectors and incidence surfaces; the five values of τ reuse the same assignments and noise; and the oracle and non-oracle models are fit to the same outcomes. Assignments and noise are independent across blocks, but blocks within an incidence configuration share its ten surfaces.

For each scenario and estimator, we computed five metrics over the valid fits, those with a finite estimate and standard error: bias, the mean of $\hat{\tau} - \tau$; the standard deviation (SD) of $\hat{\tau}$; mean squared error, $\text{MSE} = \text{mean}\{(\hat{\tau} - \tau)^2\}$; coverage, the fraction of fits with $L \leq \tau \leq U$, where $L, U = \hat{\tau} \mp 1.96 \widehat{\text{SE}}$; and power, the fraction of fits with $L > 0$, a one-sided test of $\tau = 0$ at level 0.025. For both estimators, every one of the 250 fits in every scenario was valid, and the fitting routine raised no warning other than the aliasing flags described above.

Monte Carlo standard errors come from the surface-level metrics. For a metric m , we computed m_k on the 25 fits from surface k and set $\text{SE}(m) = \text{sd}(m_1, \dots, m_{10})/\sqrt{10}$, with intervals $m \pm t_{0.975, 9} \text{SE}(m)$. A difference between two designs is paired at the surface level: its standard

error is that of $m_{d,k} - m_{d',k}$ over k . For a metric pooled over the five incidence configurations, we apply the same formula to the 50 configuration-by-surface means, each averaged over that surface’s 32 blocks; these 50 means are independent of one another. Appendix A2 gives the parameter grid, the seed keys, and the exact formulas for these metrics and standard errors.

We compared designs within each neighbor type with a Friedman rank-sum test on scenario MSE [26], treating the 160 blocks as repeated measures across the six designs. At $\tau = 1$ we followed it with Nemenyi post-hoc comparisons [27, 28] and paired Wilcoxon signed-rank tests [29] with Holm’s adjustment for multiple comparisons [30] for all 15 pairs of designs, and we repeated the Friedman test at each other value of τ . Blocks within an incidence configuration share their surfaces, so these tests treat dependent blocks as independent, which overstates the information about variation from surface to surface. For each pair of designs adjacent in the mean-MSE order, we therefore also ran a paired Wilcoxon signed-rank test over 50 independent units, the 5 configurations times 10 surfaces, with each unit’s MSE averaged over that surface’s blocks, and a t_{49} interval for the mean difference. Because the values of τ share assignments and noise, Friedman tests at different values of τ are not independent confirmations of one another.

We report results separately for each neighbor type, with queen contiguity primary, and no number is averaged over the two. Unless stated otherwise, every number in the Results is pooled over the five incidence configurations, and every table and figure caption states whether it pools the configurations or shows them separately. We report the ranking within each configuration separately and show the split wherever a conclusion changes.

Results

Design performance under queen contiguity

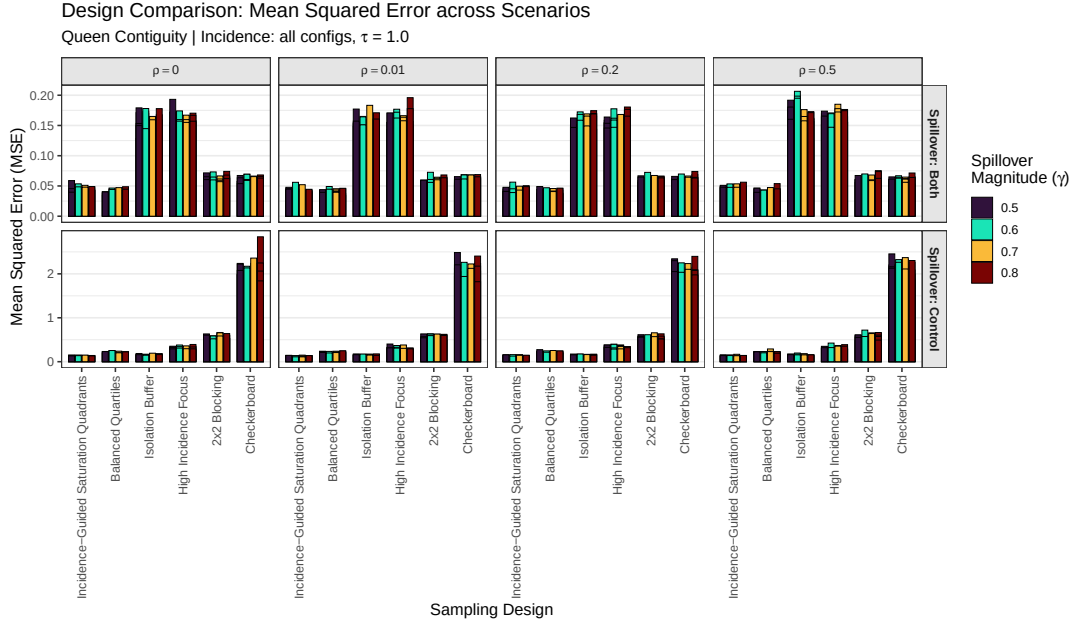
At the primary scenario, $\tau = 1$, under queen contiguity and pooled over the five incidence configurations, Incidence-Guided Saturation Quadrants had the lowest mean squared error (MSE

299 = 0.091), followed by Balanced Quartiles (0.131), Isolation Buffer (0.160), High Incidence Focus
 300 (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087) (Table 2, Figure 2a). Coverage of
 301 the 95% intervals was between 0.936 and 0.941 for all six designs (Figure 3a). Power to reject
 302 $\tau = 0$ was 0.930 for Incidence-Guided Saturation Quadrants, 0.827 for Balanced Quartiles, 0.731
 303 for Isolation Buffer, 0.642 for 2×2 Blocking, 0.610 for High Incidence Focus, and 0.547 for
 304 Checkerboard.

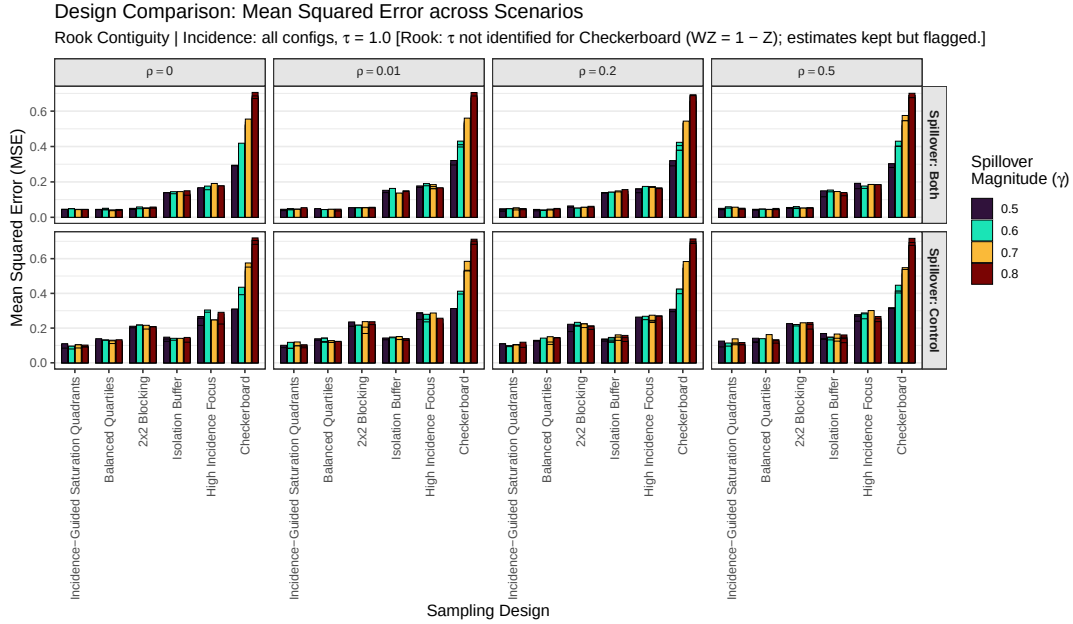
305 Incidence-Guided Saturation Quadrants also led block by block. Of the 160 queen blocks at
 306 $\tau = 1$, Incidence-Guided Saturation Quadrants had the lowest MSE in 95 (59%), Balanced
 307 Quartiles in 58 (36%), and Isolation Buffer in 7 (4%); no other design had the lowest MSE in
 308 any block.

Table 2: Performance of the six designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Monte Carlo standard errors of MSE, in parentheses, are computed from the 50 configuration-by-surface means (Methods); those of bias, coverage, and power are all below 0.007. Average rank is the mean of each design's within-block MSE rank among the six designs over the 160 blocks (1 = lowest MSE). Clusters treated is the mean number of clusters treated per assignment. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged, and its coefficient on Z estimates $\tau - \gamma$.

Design	MSE (SE)	Bias	Coverage	Power	Average rank	Clusters treated
<i>Queen contiguity (primary)</i>						
Incidence-Guided Saturation Quadrants	0.091 (0.0008)	0.032	0.937	0.930	1.425	50
Balanced Quartiles	0.131 (0.0011)	0.017	0.936	0.827	2.138	50
Isolation Buffer	0.160 (0.0011)	0.003	0.939	0.731	3.694	21.4
High Incidence Focus	0.245 (0.0047)	0.031	0.938	0.610	4.763	50
2×2 Blocking	0.319 (0.0032)	0.010	0.939	0.642	4.244	50
Checkerboard	1.087 (0.0101)	0.001	0.941	0.547	4.737	50
<i>Rook contiguity (sensitivity)</i>						
Incidence-Guided Saturation Quadrants	0.072 (0.0006)	0.018	0.942	0.963	1.488	50
Balanced Quartiles	0.084 (0.0006)	0.008	0.940	0.917	1.775	50
Isolation Buffer	0.133 (0.0009)	0.003	0.939	0.806	3.381	38.2
High Incidence Focus	0.206 (0.0030)	0.018	0.939	0.645	4.969	50
2×2 Blocking	0.128 (0.0012)	0.004	0.938	0.818	3.400	50
Checkerboard (flagged)	0.480 (0.0015)	-0.652	0.150	0.414	5.987	50

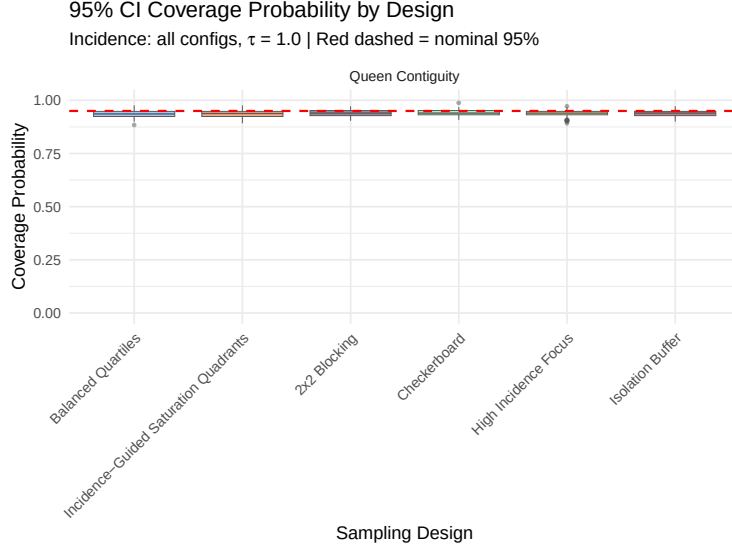


(a) Queen contiguity (primary)

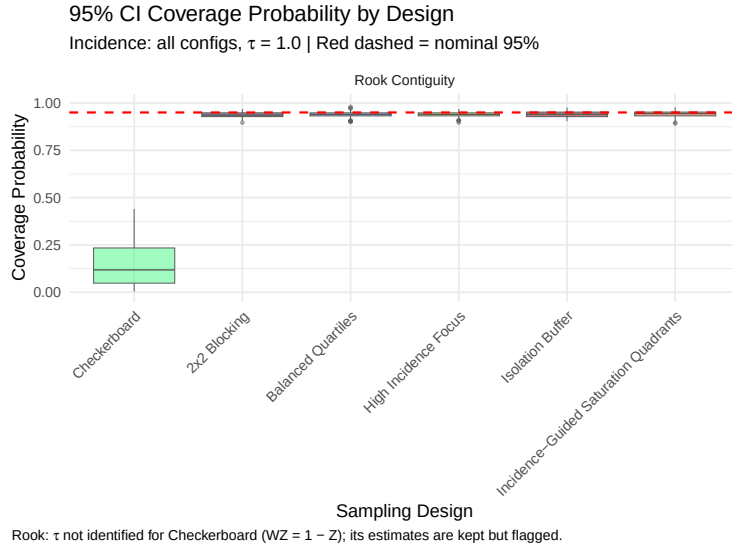


(b) Rook contiguity (sensitivity)

Figure 2: Mean squared error (MSE) of the oracle estimate of τ by design at $\tau = 1$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Columns of panels correspond to the four levels of outcome spatial dependence ρ , rows to the both-arms (top) and control-only (bottom) spillover regimes, and bar colors to the spillover magnitude γ . Each bar overlays the MSEs of the five incidence configurations for one combination of design, ρ , γ , and regime: the bar's height is the largest of the five, and the horizontal lines within it mark the other four. Designs are ordered by pooled mean MSE for the neighbor type of the panel (Table 2). The vertical scale is set separately for each row. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 3: Coverage of the 95% confidence intervals by design at $\tau = 1$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each box summarizes the coverage of the 160 scenarios of one design (5 incidence configurations $\times$ 4 values of $\rho \times$ 4 values of $\gamma \times$ 2 spillover regimes), each estimated from 250 fits. Designs are ordered by median coverage within each panel, so the order differs between panels and from the MSE order used elsewhere. The dashed red line marks the nominal 0.95 level. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Formal statistical comparison

The Friedman test rejected equal MSE ranks across the six designs under queen contiguity at $\tau = 1$ ($\chi^2 = 451.57$, $\text{df} = 5$, 160 blocks, $p < 2.2 \times 10^{-16}$). The average within-block ranks were 1.43 for Incidence-Guided Saturation Quadrants, 2.14 for Balanced Quartiles, 3.69 for Isolation Buffer, 4.24 for 2×2 Blocking, 4.74 for Checkerboard, and 4.76 for High Incidence Focus. Nemenyi comparisons (critical difference 0.596) separated 11 of the 15 pairs (Appendix A3 shows the critical-difference diagrams). Incidence-Guided Saturation Quadrants ranked significantly better than every other design, including Balanced Quartiles ($p = 0.0086$), and Balanced Quartiles ranked significantly better than the remaining four. The four pairs that Nemenyi comparisons did not separate all lay among the bottom four designs: 2×2 Blocking against Isolation Buffer ($p = 0.090$), High Incidence Focus ($p = 0.130$), and Checkerboard ($p = 0.170$), and Checkerboard against High Incidence Focus ($p = 1.00$). Holm-adjusted Wilcoxon signed-rank tests separated all 15 pairs.

The bottom three designs were ordered differently by average rank and by mean MSE. By mean MSE the order was High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087); by average rank it was 2×2 Blocking (4.24), Checkerboard (4.74), and High Incidence Focus (4.76). Both reversals have the same source: the MSEs of 2×2 Blocking and Checkerboard depended strongly on the spillover regime. Each had a lower MSE than High Incidence Focus in all 80 both-arms blocks (mean MSE 0.062 for each, against 0.162) and a higher MSE in all 80 control-only blocks (0.576 and 2.111, against 0.329; see the spillover-regime results below). Average rank counts each block once, so their wins under the both-arms regime offset their losses under the control-only regime; mean MSE weights those losses by their size.

Because blocks within an incidence configuration share their ten surfaces, we also compared each pair of designs adjacent in the mean-MSE order over the 50 independent configuration-by-surface units. Every adjacent pair differed significantly. The mean paired differences in MSE, with t_{49}

95% intervals, were -0.0401 (-0.0431 , -0.0372) for Incidence-Guided Saturation Quadrants against Balanced Quartiles, -0.0293 (-0.0332 , -0.0255) for Balanced Quartiles against Isolation Buffer, -0.0853 (-0.0954 , -0.0752) for Isolation Buffer against High Incidence Focus, -0.0738 (-0.0858 , -0.0617) for High Incidence Focus against 2×2 Blocking, and -0.7677 (-0.7893 , -0.7460) for 2×2 Blocking against Checkerboard, with Wilcoxon $p \leq 4.9 \times 10^{-9}$ in every case. The better design of each pair had the lower MSE in all 50 units, except High Incidence Focus against 2×2 Blocking, where it did so in 48 (96%). This test follows the mean-MSE order. It separated High Incidence Focus from 2×2 Blocking, which the Nemenyi comparison of within-block ranks did not, because each unit's MSE averages over both spillover regimes, and there the control-only losses of 2×2 Blocking outweigh its both-arms wins.

Design rankings by incidence configuration

Table 3 gives each design's MSE within each of the five incidence configurations. Under queen contiguity, the mean-MSE order within every configuration was the same as the pooled order: Incidence-Guided Saturation Quadrants, Balanced Quartiles, Isolation Buffer, High Incidence Focus, 2×2 Blocking, and Checkerboard. The configuration changed each design's MSE only slightly: in this order, each design's highest MSE across the five configurations was lower than the next design's lowest. Incidence-Guided Saturation Quadrants had the lowest MSE in 18 to 21 of the 32 blocks in each configuration. Coverage did not depend on the configuration either: within each configuration, every design's coverage, averaged over its 32 scenarios, was between 0.931 and 0.943. The mechanism that generated the incidence surface therefore did not change the ranking. This includes High Incidence Focus, the design that uses the incidence ranks most directly: its MSE was no higher under the two Poisson configurations than under the iid and spatial configurations.

Table 3: Mean squared error by design within each of the five incidence configurations at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity). Each entry averages the 32 scenarios of one design, neighbor type, and configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. The spatially autocorrelated and Poisson configurations are labeled by the strength of dependence in the incidence surface, ρ_X . Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	iid	Spatial		Poisson	
		$\rho_X = 0.20$	$\rho_X = 0.50$	$\rho_X = 0.20$	$\rho_X = 0.50$
<i>Queen contiguity</i>					
Incidence-Guided Saturation Quadrants	0.093	0.087	0.092	0.090	0.092
Balanced Quartiles	0.130	0.130	0.131	0.128	0.135
Isolation Buffer	0.160	0.159	0.159	0.162	0.161
High Incidence Focus	0.245	0.267	0.245	0.232	0.238
2×2 Blocking	0.311	0.315	0.323	0.326	0.320
Checkerboard	1.091	1.068	1.086	1.112	1.077
<i>Rook contiguity</i>					
Incidence-Guided Saturation Quadrants	0.072	0.072	0.073	0.073	0.069
Balanced Quartiles	0.081	0.084	0.084	0.084	0.085
Isolation Buffer	0.131	0.131	0.132	0.136	0.134
High Incidence Focus	0.205	0.213	0.200	0.202	0.211
2×2 Blocking	0.129	0.125	0.129	0.130	0.129
Checkerboard (flagged)	0.480	0.482	0.474	0.481	0.482

Sensitivity to spatial dependence, spillover magnitude, and spillover regime

Under queen contiguity, neither the strength of spatial dependence in the outcome nor the spillover magnitude changed the ranking. Across $\rho \in \{0, 0.01, 0.20, 0.50\}$, no design's MSE changed by more than 0.02 (Table 4). Across $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, the MSE of every design except Checkerboard also varied by less than 0.02, and Checkerboard's varied between 1.052 and 1.108 with no trend in γ (Table 5). Under rook contiguity, γ mattered for Checkerboard alone, for the reason given in the rook results below.

Table 4: Mean squared error by design across the four levels of outcome spatial dependence ρ , under queen contiguity at $\tau = 1$, pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\gamma \times 2$ spillover regimes). Rows are ordered by pooled mean MSE under queen contiguity (Table 2).

Design	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Incidence-Guided Saturation Quadrants	0.090	0.087	0.093	0.093
Balanced Quartiles	0.129	0.129	0.132	0.133
Isolation Buffer	0.161	0.157	0.158	0.165
High Incidence Focus	0.243	0.242	0.249	0.248
2×2 Blocking	0.318	0.320	0.314	0.324
Checkerboard	1.095	1.077	1.084	1.092

Table 5: Mean squared error by design across the four levels of spillover magnitude γ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of ρ $\times$ 2 spillover regimes). Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$
<i>Queen contiguity</i>				
Incidence-Guided Saturation Quadrants	0.092	0.089	0.092	0.089
Balanced Quartiles	0.132	0.128	0.132	0.131
Isolation Buffer	0.159	0.163	0.160	0.159
High Incidence Focus	0.244	0.249	0.248	0.241
2 $\times$ 2 Blocking	0.312	0.314	0.328	0.323
Checkerboard	1.108	1.052	1.086	1.101
<i>Rook contiguity</i>				
Incidence-Guided Saturation Quadrants	0.069	0.072	0.075	0.072
Balanced Quartiles	0.083	0.085	0.083	0.083
Isolation Buffer	0.131	0.133	0.134	0.133
High Incidence Focus	0.207	0.209	0.205	0.203
2 $\times$ 2 Blocking	0.129	0.129	0.127	0.128
Checkerboard (flagged)	0.295	0.403	0.538	0.684

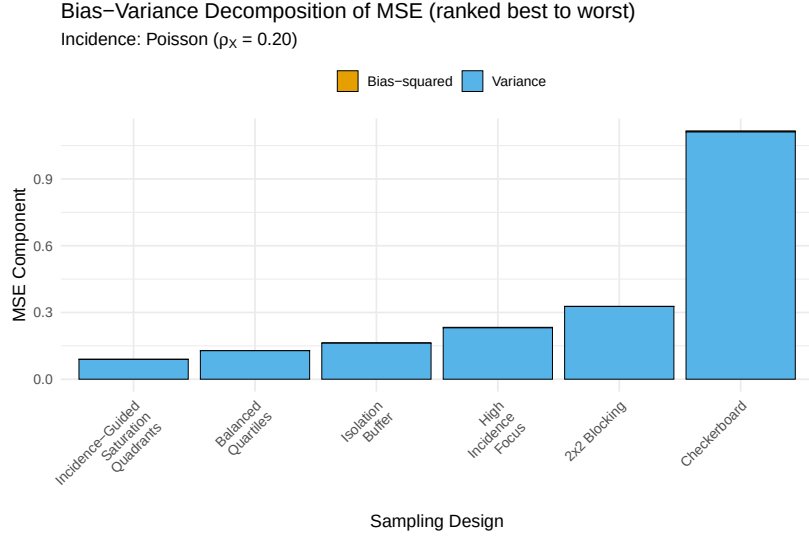
364 The spillover regime was the one factor that changed the queen ranking (Table 6, Figure 2a).
365 Under the both-arms regime, Balanced Quartiles had the lowest MSE (0.042), followed by
366 Incidence-Guided Saturation Quadrants (0.046), 2 $\times$ 2 Blocking (0.062), Checkerboard (0.062),
367 Isolation Buffer (0.161), and High Incidence Focus (0.162). Under the control-only regime, the
368 order was Incidence-Guided Saturation Quadrants (0.135), Isolation Buffer (0.159), Balanced
369 Quartiles (0.219), High Incidence Focus (0.329), 2 $\times$ 2 Blocking (0.576), and Checkerboard (2.111);
370 at larger values of τ , Isolation Buffer matched or passed Incidence-Guided Saturation Quadrants
371 under this regime (see Stability across treatment effect sizes). For Isolation Buffer, no treated
372 cluster has a treated neighbor, so $WZ = 0$ at every treated cluster and $WZ \circ (1 - Z) = WZ$. The
373 two regimes in Equation 2 therefore define the same outcome model, and so the same expected
374 MSE; the observed values, 0.161 and 0.159, differ only by Monte Carlo error, because the two
375 regimes use separate draws. Every other design had a higher MSE under the control-only regime,
376 and the increase was largest for Checkerboard and 2 $\times$ 2 Blocking, which went from close to
377 the best designs under the both-arms regime to the two worst under the control-only regime.

Incidence-Guided Saturation Quadrants was first or second under both regimes, at this and every other value of τ , which matters because a trial team may not know in advance which regime applies.

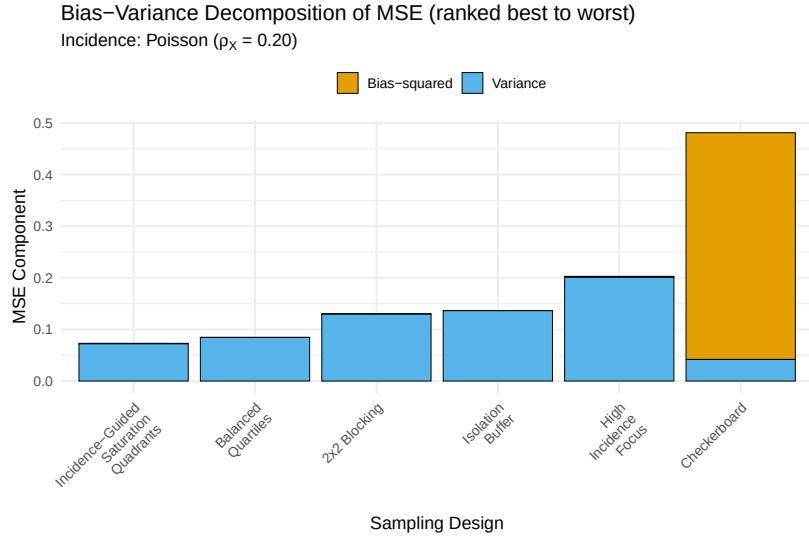
The increase under the control-only regime was variance, not bias. The oracle model includes the true regime-specific spillover covariate, so it has no omitted-variable bias. At $\tau = 1$, in every combination of design and regime under queen contiguity, the absolute bias was at most 0.048 and squared bias, averaged over scenarios, was at most 2.2% of MSE. Checkerboard showed the mechanism most clearly: under the control-only regime its spillover covariate is a linear combination of the intercept and Z at every interior cluster, so only the boundary clusters separate the spillover effect from the treatment effect (Methods). Checkerboard's MSE rose from 0.062 under the both-arms regime to 2.111 under the control-only regime, while its bias stayed near zero (0.004) and its coverage stayed at 0.942: its intervals widened with its variance. Its pooled MSE of 1.087 was almost entirely variance: squared bias, averaged over scenarios, was 0.0045, less than 0.5% of MSE (Figure 4a shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). 2×2 Blocking showed the same pattern on a smaller scale, with MSE rising from 0.062 to 0.576 and a bias of 0.016 under the control-only regime.

Table 6: Mean squared error by design under each spillover regime at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 80 scenarios (5 incidence configurations $\times$ 4 values of $\rho \times$ 4 values of γ). Rows are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	Queen		Rook	
	Both arms	Control only	Both arms	Control only
Incidence-Guided Saturation Quadrants	0.046	0.135	0.045	0.098
Balanced Quartiles	0.042	0.219	0.042	0.125
Isolation Buffer	0.161	0.159	0.131	0.134
High Incidence Focus	0.162	0.329	0.164	0.249
2×2 Blocking	0.062	0.576	0.051	0.206
Checkerboard (flagged under rook)	0.062	2.111	0.477	0.483



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 4: Decomposition of mean squared error into squared bias and variance by design at $\tau = 1$ for one incidence configuration, Poisson with $\rho_X = 0.20$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). For each design, the two stacked components are the squared bias and the variance (SD^2) of $\hat{\tau}$, each averaged over the 32 scenarios of that configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes). Designs are ordered by pooled mean MSE for the neighbor type of the panel (Table 2). The vertical scales of the two panels differ. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged, and nearly all of its MSE is squared bias.

Rook contiguity

Under rook contiguity at $\tau = 1$, pooled over the five incidence configurations, the mean-MSE order was Incidence-Guided Saturation Quadrants (0.072), Balanced Quartiles (0.084), 2×2 Blocking (0.128), Isolation Buffer (0.133), High Incidence Focus (0.206), and Checkerboard (0.480) (Table 2, Figure 2b), and the same order held within each of the five incidence configurations (Table 3). Incidence-Guided Saturation Quadrants had the lowest MSE in 97 of the 160 rook blocks (61%) and Balanced Quartiles in 60 (38%). Coverage was between 0.938 and 0.942 for every design except Checkerboard, whose coverage was 0.150 (Figure 3b); within single incidence configurations, the corresponding ranges were 0.935 to 0.946, and 0.14 to 0.15 for Checkerboard. The Friedman test rejected equal ranks ($\chi^2 = 703.76$, $df = 5$, 160 blocks, $p < 2.2 \times 10^{-16}$). Nemenyi comparisons separated 13 of 15 pairs; the two they did not separate were Incidence-Guided Saturation Quadrants against Balanced Quartiles ($p = 0.74$) and 2×2 Blocking against Isolation Buffer ($p = 1.00$). Holm-adjusted Wilcoxon tests separated every pair except 2×2 Blocking against Isolation Buffer. In the surface-level comparison, Incidence-Guided Saturation Quadrants had a lower MSE than Balanced Quartiles in 48 of the 50 units (mean difference -0.0121 ; -0.0139 , -0.0102 ; Wilcoxon $p = 9.9 \times 10^{-10}$). 2×2 Blocking and Isolation Buffer were close: 2×2 Blocking had the lower MSE in 36 of the 50 units (72%; mean difference -0.0048 ; -0.0078 , -0.0017 ; Wilcoxon $p = 0.0027$), but Isolation Buffer had the slightly better average rank (3.38 against 3.40), and the two designs traded places at some values of τ (see the results for τ below). Every other adjacent pair differed in all 50 units.

Checkerboard under rook contiguity is a different case: τ is not identified, and the coefficient on Z estimates $\tau - \gamma$ (Methods). The estimates behaved accordingly. At $\tau = 1$, Checkerboard's mean bias was -0.502 , -0.600 , -0.704 , and -0.802 at $\gamma = 0.5$, 0.6 , 0.7 , and 0.8 , while its mean standard deviation stayed between 0.20 and 0.21; squared bias, averaged over scenarios, accounted for 0.438 of its MSE of 0.480 (Figure 4b shows the decomposition for the Poisson configuration

419 with $\rho_X = 0.20$). Its coverage fell from 0.316 at $\gamma = 0.5$ to 0.027 at $\gamma = 0.8$. Because $WZ = 1 - Z$,
420 the two spillover covariates in Equation 2 coincide, $\gamma WZ \circ (1 - Z) = \gamma(1 - Z) = \gamma WZ$, so
421 the two regimes define the same outcome model and the same expected MSE; the observed
422 values, 0.477 under the both-arms regime and 0.483 under the control-only regime, differ only
423 by Monte Carlo error, because the two regimes use separate noise draws (Appendix A4). Under
424 queen contiguity, by contrast, Checkerboard’s coverage stayed between 0.939 and 0.943 at every
425 level of ρ ; under rook contiguity it was between 0.134 and 0.161 (Table 7). Rook Checkerboard
426 estimates were precise but centered on the wrong quantity, the opposite of its imprecise but
427 valid queen estimates. We keep these estimates in every ranking and flag them.

Table 7: Checkerboard’s 95% confidence interval coverage across the four levels of outcome spatial dependence ρ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\gamma \times 2$ spillover regimes). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged. The nominal level is 0.95.

Neighbor type	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Queen	0.941	0.943	0.940	0.939
Rook	0.161	0.156	0.148	0.134

428 Non-oracle sensitivity analysis

429 The non-oracle model, which omits the spillover term, was biased for every design. At $\tau = 1$,
430 and setting aside Checkerboard under rook contiguity, its bias ranged from -0.12 to -0.40 and
431 its coverage from 0.51 to 0.90 across designs and neighbor types. Under queen contiguity its bias
432 ranged from -0.123 (Incidence-Guided Saturation Quadrants) to -0.244 (2×2 Blocking) and
433 its coverage from 0.709 (2×2 Blocking) to 0.896 (High Incidence Focus); under rook contiguity
434 the most biased design other than Checkerboard was Isolation Buffer (bias -0.400 , coverage
435 0.513). For Checkerboard under rook contiguity, the non-oracle estimates were identical to the
436 oracle estimates (MSE 0.480, bias -0.652 , coverage 0.150), because once the aliased spillover
437 column is dropped the two models are the same.

438 Omitting the spillover term also removes a column that can be nearly collinear with Z , and the

non-oracle estimates were accordingly less variable for some designs. Under queen contiguity, the non-oracle MSE was lower than the oracle MSE for every design, most of all for Checkerboard (oracle 1.087, non-oracle 0.108) and 2×2 Blocking (oracle 0.319, non-oracle 0.121). Under rook contiguity it was higher for every design except 2×2 Blocking (0.128 under both models) and Checkerboard (identical). These results do not change the recommendation. Incidence-Guided Saturation Quadrants also had the lowest non-oracle MSE under both neighbor types (0.087 under queen contiguity and 0.086 under rook contiguity), and where a design's non-oracle MSE was lower, the estimate was biased and its intervals under-covered, so the lower MSE does not support valid inference.

Stability across treatment effect sizes

Under queen contiguity, each of the two design orders was the same at every $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$: the order by average within-block rank did not change across these values, and neither did the order by mean MSE, although the two orders differ from each other in the bottom three designs (see Formal statistical comparison; Figure 5a shows mean MSE). The Friedman statistic was 455.49, 451.57, 429.01, 411.19, and 376.79 at these five values ($p < 2.2 \times 10^{-16}$ at each). Under rook contiguity the order was also stable (Figure 5b) except for the close pair of 2×2 Blocking and Isolation Buffer: by average rank, 2×2 Blocking was ahead only at $\tau = 0.8$, and by mean MSE, Isolation Buffer was ahead only at $\tau = 3$. The five values of τ share their assignments and noise, so these five comparisons are not independent confirmations of one another; they show that, under queen contiguity, the order did not change across the five values of τ we tested, given the same draws.

Within that stable order, the margins moved. Under queen contiguity, the average rank of Incidence-Guided Saturation Quadrants rose from 1.41 at $\tau = 0.8$ to 1.85 at $\tau = 3$, but Balanced Quartiles did not gain on it: Incidence-Guided Saturation Quadrants had the lower MSE of the two in 103, 102, 102, 102, and 101 of the 160 blocks at $\tau = 0.8, 1, 1.5, 2$, and 3, and the average

rank of Balanced Quartiles stayed between 2.13 and 2.14. Isolation Buffer gained instead, and its average rank fell from 3.71 to 3.31. The MSE of Isolation Buffer stayed at 0.160 at every value of τ , while that of Incidence-Guided Saturation Quadrants rose from 0.088 to 0.115 and that of Balanced Quartiles from 0.130 to 0.152, so the gap between Incidence-Guided Saturation Quadrants and Balanced Quartiles stayed between 0.037 and 0.041. Isolation Buffer's gain came under the control-only regime, where Incidence-Guided Saturation Quadrants had a lower MSE than Isolation Buffer in 75, 73, 56, 40, and 12 of the 80 blocks at the five values of τ . The mean control-only MSE of Incidence-Guided Saturation Quadrants rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$, while that of Isolation Buffer stayed between 0.159 and 0.160 (Appendix A4). At $\tau = 2$ the two were level (0.1596 for Isolation Buffer against 0.1604), and at $\tau = 3$ Isolation Buffer had the lower control-only MSE (0.160 against 0.182). Incidence-Guided Saturation Quadrants remained second under the control-only regime at those two values, second to Balanced Quartiles under the both-arms regime at every value of τ , and lowest in pooled MSE at every value of τ . Its worst case across blocks was the lowest of the six designs at $\tau \leq 1.5$, but at $\tau = 2$ and $\tau = 3$ that of Isolation Buffer was lower (0.207 against 0.210, and 0.208 against 0.238). The mean bias of Incidence-Guided Saturation Quadrants also grew with τ , from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$, but squared bias, averaged over scenarios, stayed at or below 3.0% of its MSE, so the rise in its MSE was mostly variance.

Under rook contiguity, Incidence-Guided Saturation Quadrants kept the lowest control-only MSE at every value of τ , although its margin over Isolation Buffer narrowed, from 0.095 against 0.134 at $\tau = 0.8$ to 0.131 against 0.136 at $\tau = 3$, where it had the lower MSE in 47 of the 80 control-only blocks. Its worst case across rook blocks was the lowest of the six designs at every value of τ except $\tau = 3$, where that of Isolation Buffer was lower (0.173 against 0.174).

Except for Checkerboard under rook contiguity, coverage stayed close to 0.94 at every value of τ . Pooled over the five incidence configurations, every design's coverage was between 0.936

and 0.941 at every value of τ under queen contiguity, and every design except Checkerboard was between 0.937 and 0.942 at every value of τ under rook contiguity. The coverage of rook Checkerboard, for which τ is not identified, rose from 0.138 at $\tau = 0.8$ to 0.150, 0.198, 0.269, and 0.442 at $\tau = 1, 1.5, 2$, and 3 (Appendix A4). Appendix A3 shows coverage across τ for each design.

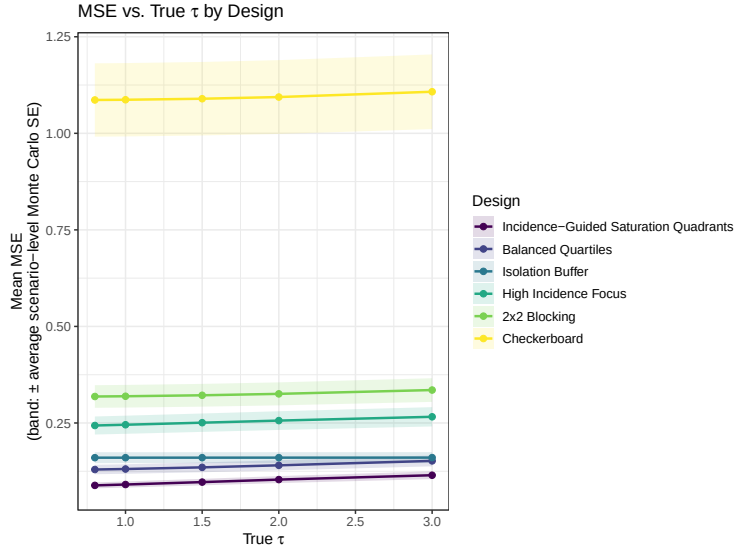
Robustness across blocks and the recommended design

Table 8 gives the distribution of MSE across the 160 queen blocks at $\tau = 1$. Incidence-Guided Saturation Quadrants had the lowest median (0.080) and the lowest worst-case MSE of any design (0.169). Isolation Buffer had the narrowest range (0.128 to 0.206), because its MSE does not depend on the spillover regime, but at a higher level. Balanced Quartiles had a low median (0.115) and a worst case of 0.290. The distributions of 2×2 Blocking and Checkerboard reflected their dependence on the spillover regime: both had low first quartiles (0.062 and 0.063) and high upper quartiles (0.580 and 2.093). Checkerboard had the highest MSE in all 80 control-only blocks; in the 80 both-arms blocks, the highest MSE belonged to High Incidence Focus in 42 and to Isolation Buffer in 38. Under rook contiguity, Incidence-Guided Saturation Quadrants also had the lowest worst-case MSE (0.138).

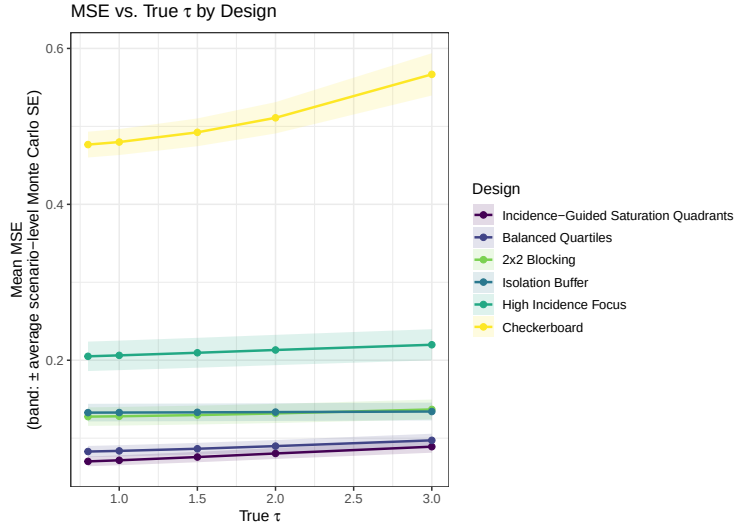
Table 8: Distribution of block-level mean squared error by design across the 160 blocks under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (32 blocks in each). At a fixed value of τ , each block contributes one scenario per design. Rows are sorted by median, which here coincides with the order of pooled mean MSE under queen contiguity (Table 2).

Design	Best	Q25	Median	Q75	Worst
Incidence-Guided Saturation Quadrants	0.037	0.045	0.080	0.135	0.169
Balanced Quartiles	0.034	0.042	0.115	0.221	0.290
Isolation Buffer	0.128	0.150	0.159	0.169	0.206
High Incidence Focus	0.139	0.160	0.234	0.322	0.424
2×2 Blocking	0.048	0.062	0.271	0.580	0.719
Checkerboard	0.051	0.063	0.899	2.093	2.843

We recommend Incidence-Guided Saturation Quadrants. Under both neighbor types, it had the lowest mean MSE, pooled over the two spillover regimes, in every incidence configuration



(a) Queen contiguity (primary)



Rook: τ not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept but flagged.

(b) Rook contiguity (sensitivity)

Figure 5: Mean squared error by design across the five values of the treatment effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each point averages the 160 scenarios of one design at one value of τ , pooled over the five incidence configurations. Shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of MSE; they describe the precision of a single scenario's MSE, not of the plotted average, whose standard error is smaller (Table 2). The five values of τ share their assignments and noise. The vertical scales of the two panels differ. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

and at every value of τ , and it was first or second under each spillover regime at every value of τ . At $\tau = 1$, its worst case across blocks was the lowest of the six designs, and its intervals had coverage of 0.937 under queen contiguity and 0.942 under rook contiguity. Under queen contiguity, Isolation Buffer, whose MSE does not depend on the regime, matched it under the control-only regime at $\tau = 2$ and passed it at $\tau = 3$, but was fifth of the six designs under the both-arms regime at every value of τ (fourth under rook contiguity). Balanced Quartiles also performed well. It was second overall under both neighbor types at every value of τ , and it was better than Incidence-Guided Saturation Quadrants only under the both-arms regime. At $\tau = 1$ that advantage was marginal (0.0421 against 0.0459 under queen contiguity, 0.0421 against 0.0451 under rook contiguity), and Incidence-Guided Saturation Quadrants had the lower MSE in 22 of 80 queen blocks and 23 of 80 rook blocks. Under the control-only regime, Incidence-Guided Saturation Quadrants had the lower MSE in all 80 queen blocks and 75 of 80 rook blocks, by a wider margin (0.135 against 0.219 under queen contiguity).

Application: A Planned Sudden Unexpected Death Prevention

Pilot in North Carolina

Background

SUD in adults is an incompletely understood cause of premature mortality. The SUDEN (Sudden Unexpected Death in North Carolina) methodology, a validated algorithm for classifying deaths as sudden and unexpected from death certificate and medical record data, was developed and screened against expert physician panel review by Nanavati and colleagues [31]. Using this methodology, Mirzaei and colleagues estimated SUD incidence rates of 49.3 per 100,000 among US men and 21.7 per 100,000 among US women aged 20–64, extrapolated from Wake County, North Carolina (see Heterogeneous baseline incidence). They found that lifetime productivity loss from SUD exceeded that from any individual cancer, and that among natural causes of death

only all cancers combined and heart disease accounted for more years of life lost than SUD [24]. Subsequent work has documented geographic and socioeconomic gradients in SUD incidence within North Carolina: Gan and colleagues found elevated SUD incidence in rural counties relative to urban counties [32], and Mounsey and colleagues found an inverse relationship between household income and SUD incidence in Wake County [33]. SUD is therefore a measurable outcome in North Carolina whose incidence varies spatially and socioeconomically, which makes it relevant to the incidence-informed designs evaluated in this chapter.

The planned pilot design

A pilot intervention is being planned in collaboration with North Carolina’s community college (CC) system, which provides the spatial and organizational unit for the design framework developed in this chapter. North Carolina’s 58 community colleges collectively serve all 100 counties in the state; each college has a primary campus in one county and, in many cases, satellite campuses in neighboring counties (for example, Durham Technical Community College operates a satellite campus in Orange County in addition to its primary Durham County campus). For this application, each county is assigned to exactly one CC service area, so the 58 service areas partition the 100 counties. Northampton and Bertie counties also lie within Roanoke-Chowan Community College’s broader service network; we assign Northampton County to Halifax Community College, Bertie County to Martin Community College, and Hertford County alone to Roanoke-Chowan Community College. This mapping is a convention of this study’s design, not a statement about the colleges’ service commitments. Figure 6 shows the resulting partition of the 100 counties into the 58 service areas. This network structure motivates the spillover term in this chapter’s outcome model. Community colleges within a shared service network communicate and coordinate programmatic activity with one another, so an intervention delivered at one college plausibly influences outcomes in a neighboring college’s service area even without any formal expansion of the intervention itself. The spillover term in Equation 2

approximates this mechanism: it passes spillover between adjacent clusters through the contiguity weights W , whereas coordination between colleges follows their programmatic ties, which need not coincide with shared service-area boundaries.

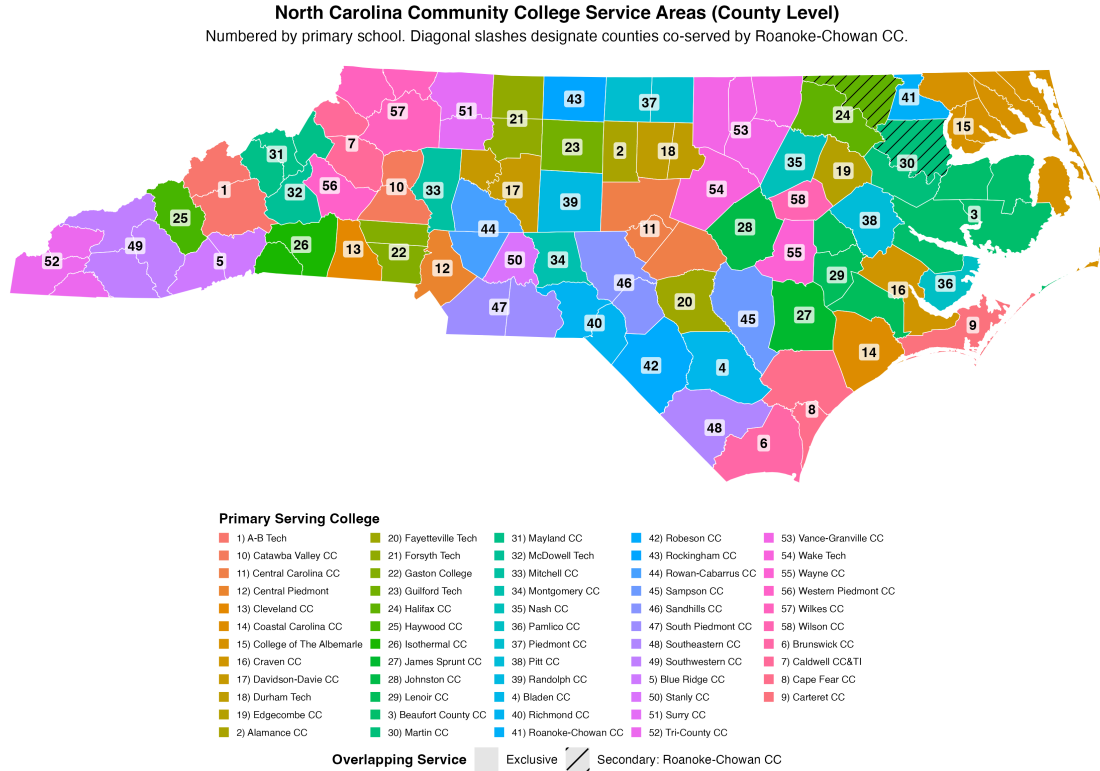


Figure 6: The assignment of North Carolina’s 100 counties to the 58 community college service areas used as clusters in the planned application. Each county is colored by its primary serving college, and each service area carries a single number, keyed to the legend. Hatching marks the two counties that also lie within Roanoke-Chowan Community College’s broader service network: Northampton County, assigned to Halifax Community College (24), and Bertie County, assigned to Martin Community College (30). Roanoke-Chowan Community College (41) serves Hertford County alone in this analysis. The planned pilot site, Lenoir Community College, is service area 29. The map shows county boundaries and the county-to-college assignment only; it does not depend on the SUD data.

The planned pilot site is Lenoir Community College, whose service area comprises Lenoir, Greene, and Jones counties. The pilot cluster is that whole service area, with anticipated spillover to adjacent CC service areas. The pilot intervention involves workforce development and disaster-response-style training intended to teach CC students and staff to recognize signs of medical distress, connect at-risk individuals to care, and support timely follow-up. The teaching intervention under consideration is delivered county-wide by design, rather than at finer spatial

resolution such as the census tract level, so the treatment unit for this application is the CC service area (equivalently, a group of one or more counties) rather than the census tract. Individual-level incidence data may still be collected at finer resolution to inform the incidence-aware design components. The eventual evaluation plan anticipates that the intervention’s effect will be visible first in intermediate process measures, such as medication adherence and the timeliness of clinic visits following an at-risk event, since meaningful shifts in the SUD incidence rate itself are not expected to be detectable on the timescale of an initial pilot. This expected measurement lag between process outcomes and the ultimate incidence outcome modeled throughout this chapter is discussed further as a limitation below.

County-level SUD incidence for 2018–2021 was derived by Habib from North Carolina death certificates [34]. Of the 412,514 deaths recorded statewide over that period, that derivation kept the 111,665 deaths among working-age adults (aged 18–64). It excluded deaths in hospitals and other institutions and deaths from expected causes (such as trauma, overdose, COVID-19, and end-stage chronic disease), and it applied a death-certificate algorithm validated against the SUDDEN physician-adjudicated cohort [32, 31]. The result is 21,147 sudden unexpected out-of-hospital deaths among working-age adults, pooled to the county level using death-certificate zip codes, with population-at-risk denominators from Surveillance, Epidemiology, and End Results (SEER) population estimates for ages 18–64 in the corresponding years. We have aggregated these county-level data to the 58 community college service-area clusters and computed cluster-level incidence for each year from 2018 to 2021 (see Data preparation and planned analysis). Applying the six designs to these incidence surfaces and comparing them is the next step and has not yet been done. Planned covariates for the eventual real-data application, once finalized, include county-level median income, food-desert status, physician access, and average driving distance to the nearest hospital.

Data preparation and planned analysis

The analysis has five steps, and the first three are complete. First, we aggregated the county-level SUD death counts and SEER populations aged 18–64 to the 58 CC service-area clusters, assigning each county to its single cluster. Second, we computed cluster-level SUD incidence for each year from 2018 to 2021, as the cluster’s deaths in that year divided by its population aged 18–64 in that year, per 100,000, and a pooled 2018–2021 rate per 100,000 person-years, as the cluster’s deaths summed over the four years divided by its population summed over the four years. The aggregated data reproduce the statewide death count, person-years, and yearly and overall rates reported by Habib [34], and in every year the 58 cluster totals sum to the statewide totals. Third, we built queen and rook contiguity weights for the 58 clusters, row-standardized as in the grid study; queen contiguity matches the primary analysis of this chapter. The weights use legal county boundaries rather than shoreline-clipped ones, so clusters whose counties meet across a river or sound are neighbors, because an education intervention can plausibly reach nearby counties across water. Fourth, we will apply the six designs to the 58 clusters. The recommended design, Incidence-Guided Saturation Quadrants, is defined on the quadrants of a regular grid, so for the service-area geography its four quadrants are replaced by four regions: k-means clustering of the service areas’ locations generates 250 candidate partitions, and the partition chosen is the one that best balances population and the numbers of clusters and counties across regions while keeping each region compact. The regions are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean, and within each region that share of its clusters is treated at random. The existing application code, which predates the revised simulation and is to be finalized, also adapts three other designs to the service-area geography: Checkerboard becomes a greedy two-coloring of the contiguity graph, in which each cluster, taken in decreasing order of its number of neighbors, is given the arm held by fewer of its already-assigned neighbors; 2×2 Blocking becomes k-means blocks of about four clusters, formed from the clusters’ locations, incidence, and population, with half of each

block, rounded, treated at random; and High Incidence Focus treats the 29 highest-incidence clusters, half of the 58. Fifth, we will compare the designs on the real cluster-level incidence surface, using the same simulation machinery as the grid-based comparison. The existing code for the comparison on the service-area geography predates the revised simulation described in Methods and has not yet been run on these data.

Application results (to be completed)

The results of this analysis will be reported here once it has been run.

Discussion

Summary and relevance

We compared six treatment assignment designs for a spatial cluster randomized trial on a grid of 100 clusters, under three mechanisms of heterogeneous baseline incidence, two spillover regimes, and a range of values for spatial dependence, spillover magnitude, and the treatment effect. Some findings held across the incidence configurations and the values of ρ , γ , and τ under queen contiguity. Others depended on the spillover regime, which a trial team may not know in advance.

Under queen contiguity, Incidence-Guided Saturation Quadrants had the lowest mean squared error (0.091) and Balanced Quartiles the second lowest (0.131). The order of the six designs was the same in each of the five incidence configurations and at every value of τ , and neither the strength of spatial dependence in the outcome nor the spillover magnitude changed it. The mechanism that generated the incidence surface changed each design's MSE only slightly. The same two designs also led under rook contiguity (0.072 and 0.084). Coverage of the 95% intervals was close to 0.94 for every design under both neighbor types, except Checkerboard under rook contiguity, so the designs differed in precision and coverage did not separate them.

Under queen contiguity, Checkerboard and 2×2 Blocking were close to the best designs when spillover reached both arms and were the two worst when it reached only control clusters: at $\tau = 1$, Checkerboard's MSE was 0.062 under the both-arms regime and 2.111 under the control-only regime, and that of 2×2 Blocking was 0.062 and 0.576. Balanced Quartiles also lost ground under the control-only regime (0.042 and 0.219), while Incidence-Guided Saturation Quadrants was first or second under both regimes at every value of τ . Its control-only MSE did rise with τ , and Isolation Buffer, whose MSE did not, matched it under that regime at $\tau = 2$ and passed it at $\tau = 3$. A design whose performance depends on the regime carries a risk that its average MSE does not show.

Checkerboard failed for different reasons under the two neighbor types. Under queen contiguity, its high MSE was variance, not bias: under the control-only regime, only the boundary clusters separate the spillover effect from the treatment effect (Methods). The estimates stayed centered on τ (bias 0.004 under the control-only regime at $\tau = 1$) and their intervals kept close to nominal coverage (0.942), but they were imprecise. Under rook contiguity, τ is not identified (Methods): the estimates were centered on $\tau - \gamma$, so rook Checkerboard's MSE grew with γ , and its coverage at $\tau = 1$ was 0.150. Checkerboard intersperses treated and control clusters as finely as the grid allows; both failures come from the dependence this pattern creates between Z and the spillover covariate, which is a linear combination of the intercept and Z at every cluster under rook contiguity and at every interior cluster under queen contiguity and the control-only regime.

We recommend Incidence-Guided Saturation Quadrants for the reasons given in the Results, chiefly that, under both neighbor types, it had the lowest pooled MSE and was first or second under each spillover regime at every value of τ , and that at $\tau = 1$ its worst case across blocks was the lowest of the six designs. Balanced Quartiles also worked well; its advantage under the both-arms regime was small, and its disadvantage under the control-only regime was large.

The saturation design did as well when its saturations were assigned without reference to

incidence (Saturation Quadrants; see Treatment assignment designs). Both designs treat the four quadrants at saturations from 20% to 80%, but no design in this chapter removes that structure while keeping the rest fixed, so these results do not show that the saturation structure is the reason for their lower MSE.

These are design-stage results, and they complement analysis-stage modeling rather than replace it. Every design was analyzed with a spatial lag model that included the spillover term, and for every design whose τ was identified the estimates were close to unbiased: at $\tau = 1$, the absolute bias of each design under each regime was at most 0.048 under queen contiguity and 0.027 under rook contiguity. Given that model, the choice of design determined how precisely τ was estimated.

Relation to Chapter 2

Checkerboard is the strict allocation of the block-stratified design that Chapter 2 recommended over simple random assignment on rook grids of 8 to 12 clusters. Chapter 2 used the term “checkerboard paradox” for the amplified bias it observed for this design on its 3×4 grid when spillover reached both arms and the outcome was spatially dependent. In its limitations, it also noted that an allocation in which treatment and spillover exposure are exactly collinear gives $\hat{\beta} \approx \beta - \psi$ in its notation. A rook checkerboard is such an allocation on a rectangular grid of any size, and on 100 clusters its coefficient on Z estimated $\tau - \gamma$ in the same way. Under queen contiguity, a neighbor structure that Chapter 2 listed as an extension, τ is identified, and Checkerboard performed close to the best designs under the both-arms regime and worst under the control-only regime. Chapter 2 compared the design only with simple random assignment; against the wider set of designs studied here, Checkerboard had the highest mean MSE under both neighbor types.

686 Limitations

687 These conclusions have several limitations. First, the primary estimator is an oracle. It includes
688 the exact spillover exposure term, so its results are a best case for each design. We fit a
689 non-oracle model that omits the spillover term as a sensitivity analysis. At $\tau = 1$, setting aside
690 Checkerboard under rook contiguity, it was biased for every design, with bias between -0.12
691 and -0.40 and coverage between 0.51 and 0.90 , and Incidence-Guided Saturation Quadrants
692 still had the lowest MSE under both neighbor types. The non-oracle model is only one departure
693 from the oracle. We did not study an analyst who includes a spillover term of the wrong form,
694 such as one built on the wrong neighbor definition or a distance decay.

695 Second, the designs and the fitted model used the incidence surface that generated the outcome
696 (the matched-surface assumption in Methods). In practice, planning incidence is estimated from
697 an earlier period and measured with error. The designs that assign treatment by incidence rank,
698 and the covariate adjustment, would then use a noisy surface, and we did not study how that
699 changes the comparison.

700 Third, in the Poisson mechanism every cluster had the same population of 100,000. Real clusters,
701 including the community college service areas in the Application, differ in population, so their
702 observed rates differ in precision, and unequal populations may change how the designs that use
703 incidence perform.

704 Fourth, all results come from a single regular 10×10 grid, with β and σ fixed at 1. We did not
705 vary the grid's size or shape, and some results depend on its geometry, including the number of
706 clusters Isolation Buffer treats and the 36 boundary clusters that identify τ for Checkerboard
707 under queen contiguity and the control-only regime.

708 Fifth, the simulation shares random draws by design. Blocks within an incidence configuration
709 share its ten incidence surfaces, so the block-level tests treat dependent blocks as independent,
710 which is why we also compared adjacent designs over the 50 independent configuration-by-surface

units. The five values of τ also share their draws, so the stability of the design order across τ is not an independent replication at each value (see Stability across treatment effect sizes).

Sixth, the simulation's base rate of 35 per 100,000 is not the rate of the county data described in the Application (see Heterogeneous baseline incidence). We kept it deliberately, because the simulation is a general design study, not a calibration to one data set. In the Poisson mechanism, incidence enters the designs and the fitted model only through the ranks of the observed rates, so the base rate affects the simulation only through the spatial pattern of those ranks and the number of tied rates.

Seventh, in the planned application, changes in SUD incidence are not expected to be detectable on the timescale of an initial pilot, so its evaluation will rely on intermediate process outcomes. We evaluated the designs on a continuous cluster-level outcome generated by the model in Equation 1 and did not study how their performance carries over to a process outcome.

Future work

Four extensions follow from these limitations. First, a data-generating process in which the planning incidence is a noisy snapshot of a latent risk surface, and the outcome depends on the latent surface, would test the matched-surface assumption directly and show how the designs that assign treatment by incidence rank behave when those ranks are partly wrong. Second, unequal cluster populations in the Poisson mechanism would bring the simulation closer to the Application. Third, varying the size and shape of the grid, including irregular geographies such as the community college service areas, would show how much of the ranking depends on a regular 10×10 lattice. Fourth, and most immediately, applying the designs to the county-level SUD data described in the Application, now that those data are available, will show whether the recommendation holds under observed incidence.

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Appendix

A1. The eight-design comparison

Table 9 defines the two designs that we implemented but do not compare in the body of this chapter, in the format of Table 1, and Table 10 reports the comparison of all eight designs at the primary scenario, $\tau = 1$.

Table 9: The two designs implemented but not compared in the body of this chapter, in the format of Table 1. Saturation Quadrants shares its mechanism with Incidence-Guided Saturation Quadrants, and Balanced Halves with Balanced Quartiles (see Treatment assignment designs). A cluster's incidence rank is its rank by the baseline incidence surface X_k , with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment.

Design	Assignment rule	Clusters treated	Deterministic
Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants, which are given saturations of 20%, 40%, 60%, and 80% in an order drawn at random for each assignment, without reference to incidence; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Halves	Clusters are divided by incidence rank into two strata of 50; 25 clusters are treated at random in each stratum	50	No

The argument for leaving Saturation Quadrants and Balanced Halves out of the body, with the Nemenyi and Wilcoxon results for each pair, is given under Treatment assignment designs in Methods; Table 10 supplies the rest of the eight-design comparison. Under both neighbor types, each excluded design is adjacent to its retained counterpart in mean MSE. The MSEs and coverages of the six retained designs are those of Table 2, given here to four and three decimal places.

A2. Simulation mechanics

This section gives the mechanical details behind Simulation design, resampling, and metrics (Methods): the parameter grid, the random-number streams, the Monte Carlo structure, the scripts, and the exact formulas for the metrics and their Monte Carlo standard errors.

Table 10: Performance of all eight designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Retained marks the six designs compared in the body of this chapter. Average rank is the mean of each design’s within-block MSE rank among the eight designs over the 160 blocks (1 = lowest MSE), so it differs from the six-design average rank in Table 2. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	Retained	MSE	Coverage	Average rank
<i>Queen contiguity (primary)</i>				
Saturation Quadrants	No	0.0900	0.936	2.100
Incidence-Guided Saturation Quadrants	Yes	0.0907	0.937	2.350
Balanced Halves	No	0.1299	0.936	3.312
Balanced Quartiles	Yes	0.1308	0.936	3.356
Isolation Buffer	Yes	0.1602	0.939	5.138
High Incidence Focus	Yes	0.2455	0.938	6.763
2 $\times$ 2 Blocking	Yes	0.3192	0.939	6.244
Checkerboard	Yes	1.0869	0.941	6.737
<i>Rook contiguity (sensitivity)</i>				
Saturation Quadrants	No	0.0708	0.940	2.281
Incidence-Guided Saturation Quadrants	Yes	0.0717	0.942	2.306
Balanced Halves	No	0.0842	0.940	3.038
Balanced Quartiles	Yes	0.0837	0.940	2.919
Isolation Buffer	Yes	0.1329	0.939	5.225
High Incidence Focus	Yes	0.2062	0.939	6.969
2 $\times$ 2 Blocking	Yes	0.1281	0.938	5.275
Checkerboard (flagged)	Yes	0.4799	0.150	7.987

Parameter grid. Table 11 lists the parameters that were crossed to form the 12,800 scenarios and the settings that were held fixed.

Random-number streams. Every random draw in the simulation is preceded by the R call

```
set.seed(digest::digest2int(key), kind = "Mersenne-Twister",
        normal.kind = "Inversion", sample.kind = "Rejection")
```

where `digest2int` is the string-to-integer hash function of the R package `digest`, and `key` joins the draw’s key fields with the character `|`, each numeric field written with two decimal places (`sprintf("%.2f")`). Table 12 lists the three kinds of draw and their keys. Setting the three kinds explicitly fixes the Mersenne-Twister generator, the inversion method for normal draws, and rejection sampling for random sampling, so the streams do not depend on the defaults of

Table 11: Parameter grid of the full simulation. The crossed parameters form $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12,800$ scenarios, 9,600 of them for the six designs compared in the body of this chapter; the settings in the lower panel are the same in every scenario.

Parameter	Values	Levels
<i>Crossed</i>		
Treatment effect τ	0.8, 1.0 (primary), 1.5, 2.0, 3.0	5
Incidence configuration	iid; spatially autocorrelated, $\rho_X \in \{0.20, 0.50\}$; Poisson, $\rho_X \in \{0.20, 0.50\}$	5
Neighbor type	Queen (primary), rook (sensitivity)	2
Outcome spatial dependence ρ	0, 0.01, 0.20, 0.50	4
Spillover magnitude γ	0.5, 0.6, 0.7, 0.8	4
Spillover regime	Both arms, control only (Equation 2)	2
Design	The six designs of Table 1 and the two of Table 9	8
<i>Fixed</i>		
Grid	Regular 10×10 grid, $N = 100$ clusters	
Outcome model (Equation 1)	$\beta = 1$, $\sigma = 1$	
Poisson mechanism	Base rate 35 per 100,000; $P = 100,000$ per cluster	
Incidence surfaces per configuration	$K = 10$	
Assignments per surface	$J = 25$ (Checkerboard: one assignment, used in every draw)	
Fits per scenario and estimator	$KJ = 250$, each with its own noise vector	
Estimators	Oracle (primary) and non-oracle (sensitivity), both fit to every simulated outcome	

the R session. The consequences of this scheme, independence from the order of execution and common random numbers, are described in Methods.

Table 12: Seed keys. Mode is the incidence mechanism (iid, spatial, or Poisson); for the iid configuration, which has no ρ_X , that field is 0. The regime field is control only or both arms, k indexes the incidence surface, and d identifies the design. A draw is shared by every scenario whose key fields match its own.

Draw	Key fields	Shared by
Incidence surface X_k	“X”, mode, ρ_X , k	Every block, design, and value of τ of the configuration
The $J = 25$ assignments of design d on surface k	“Z”, mode, ρ_X , neighbor type, ρ , γ , regime, k , d	All five values of τ and both estimators
The 250 noise vectors of a block	“eps”, mode, ρ_X , neighbor type, ρ , γ , regime	All eight designs, all five values of τ , and both estimators

Monte Carlo structure and checkpoints. Within a scenario, fit (k, j) , with $k = 1, \dots, 10$ and $j = 1, \dots, 25$, uses surface X_k , assignment Z_{kj} , and noise vector ϵ_{kj} , the (k, j) column of the block’s 100×250 noise matrix (see Simulation design, resampling, and metrics). The unit of computation is a combination of incidence configuration, neighbor type, and ρ : 40 units, each covering 4 values of γ , 2 spillover regimes, 8 designs, 5 values of τ , 250 fits, and 2 estimators.

849 The units ran in parallel in forked R processes, each limited to a single BLAS thread, and
 850 the run stops if any unit fails. Each unit's results are saved as a checkpoint, together with a
 851 manifest that records hashes of the parameter settings and of the five simulation scripts (01–05
 852 below), the versions of the R packages `spatialreg`, `spdep`, `dplyr`, and `digest`, the R version,
 853 and the BLAS library. If any of these differs from the manifest, the runner refuses to load the
 854 checkpoints, so results from different code or settings are never combined.

855 **Scripts.** The simulation and the summaries in this chapter come from these R scripts:

- 856 • `01_spatial_setup.R` builds the 10×10 grid, its rook and queen neighbor lists, and the
 857 row-standardized weights matrices W .
- 858 • `02_incidence_generation.R` generates incidence surfaces under the iid, spatially auto-
 859 correlated, and Poisson mechanisms.
- 860 • `03_designs.R` draws assignments from the eight designs, breaking ties in incidence at
 861 random in every draw.
- 862 • `04_estimation.R` fits the oracle and non-oracle spatial lag models by maximum likelihood,
 863 with our own implementation of the estimator and, for the cross-check, with `lagsarlm`.
- 864 • `05_run_simulation.R` sets the parameters, seeds and makes every draw, fits both models,
 865 computes the metrics and their surface-level standard errors, and writes scenario-level and
 866 surface-level results for each estimator.
- 867 • `06_visualizations.R` holds shared plotting helpers, including `plot_design_samples()`,
 868 which draws Figure 1.
- 869 • `12_six_design_statistical_comparisons.R` runs the Friedman, Nemenyi, and
 870 Wilcoxon comparisons of the six designs for each neighbor type and draws the coverage
 871 and τ -sensitivity figures.
- 872 • `13_dissertation_results_extract.R` summarizes the results by incidence configuration,
 873 parameter level, and block, and runs the paired comparisons over the 50 configuration-by-

surface units.

- `14_manuscript_supplement_figures.R` runs the eight-design comparison in Appendix A1 and draws the MSE-by-design and bias-variance figures and the critical-difference and coverage-across- τ figures in Appendix A3.

Scenario metrics. For one scenario and estimator, let $f = 1, \dots, n$ index the valid fits, those with a finite estimate and standard error (here every fit was valid, so $n = 250$). Let $L_f, U_f = \hat{\tau}_f \mp z_{0.975} \widehat{\text{SE}}_f$, where $z_{0.975} = \Phi^{-1}(0.975) \approx 1.96$, and let $\bar{\tau} = n^{-1} \sum_f \hat{\tau}_f$. Then

$$\text{Bias} = \frac{1}{n} \sum_{f=1}^n (\hat{\tau}_f - \tau), \quad \text{SD} = \left\{ \frac{1}{n-1} \sum_{f=1}^n (\hat{\tau}_f - \bar{\tau})^2 \right\}^{1/2}, \quad \text{MSE} = \frac{1}{n} \sum_{f=1}^n (\hat{\tau}_f - \tau)^2,$$

$$\text{Coverage} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}\{L_f \leq \tau \leq U_f\}, \quad \text{Power} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}\{L_f > 0\},$$

so that $\text{MSE} = \text{Bias}^2 + (n-1) \text{SD}^2/n$.

Surface-level standard errors. For $m \in \{\text{Bias}, \text{MSE}, \text{Coverage}, \text{Power}\}$, let m_k be the same formula applied to the valid fits from surface k (here all 25). With $K = 10$ and $s(a_1, \dots, a_K) = \{(K-1)^{-1} \sum_k (a_k - \bar{a})^2\}^{1/2}$ the sample standard deviation,

$$\text{SE}(m) = \frac{s(m_1, \dots, m_K)}{\sqrt{K}}, \quad m \pm t_{0.975, K-1} \text{SE}(m).$$

Because every fit was valid, each surface contributes 25 fits and $m = K^{-1} \sum_k m_k$. For two designs d and d' in the same block at the same τ , the difference is paired at the surface level: with $\Delta_k = m_{d,k} - m_{d',k}$,

$$\text{SE}(m_d - m_{d'}) = \frac{s(\Delta_1, \dots, \Delta_K)}{\sqrt{K}},$$

with the same $t_{0.975, K-1}$ interval.

Standard errors pooled over configurations. For one design, neighbor type, and value of τ , let $\mathcal{B}_c$ be the 32 blocks of configuration c and $m_{b,k}$ the surface- k metric in block b . The

892 configuration-by-surface means and the pooled metric are

$$\bar{m}_{c,k} = \frac{1}{32} \sum_{b \in \mathcal{B}_c} m_{b,k}, \quad \bar{m} = \frac{1}{50} \sum_{c=1}^5 \sum_{k=1}^{10} \bar{m}_{c,k},$$

893 where $\bar{m}$ equals the mean of the metric over the 160 scenarios, and

$$\text{SE}(\bar{m}) = \frac{s(\{\bar{m}_{c,k}\})}{\sqrt{50}},$$

894 with s taken over the 50 values $\bar{m}_{c,k}$. These are the standard errors in Table 2. For two designs d
895 and d' adjacent in the mean-MSE order (Formal statistical comparison), let $D_{c,k} = \bar{m}_{d,c,k} - \bar{m}_{d',c,k}$
896 for $m = \text{MSE}$. The interval for the mean difference is $\bar{D} \pm t_{0.975, 49} s(\{D_{c,k}\})/\sqrt{50}$, and the
897 Wilcoxon signed-rank test, with the normal approximation, is applied to the 50 values $D_{c,k}$.

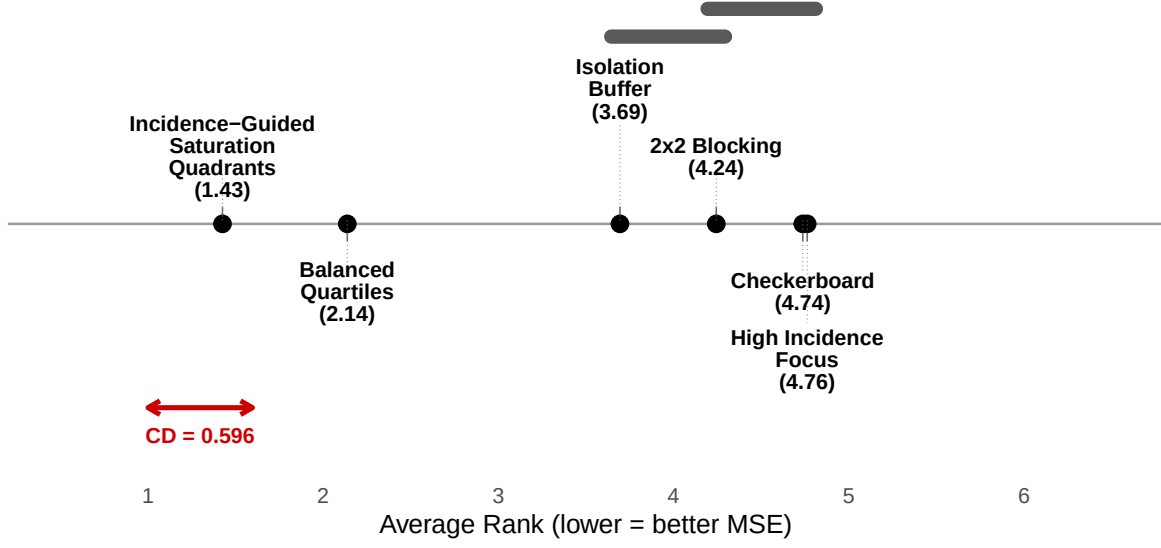
898 **A3. Supplementary statistical comparisons**

899 This section shows two figures behind results stated in the body of this chapter. Figure 7 gives
900 the Nemenyi critical-difference diagrams for the comparison of the six designs at $\tau = 1$, whose
901 results are reported under Formal statistical comparison (queen contiguity) and Rook contiguity
902 in the Results. Figure 8 gives the coverage of each design across the five values of τ , summarized
903 under Stability across treatment effect sizes.

904 **A4. Technical notes**

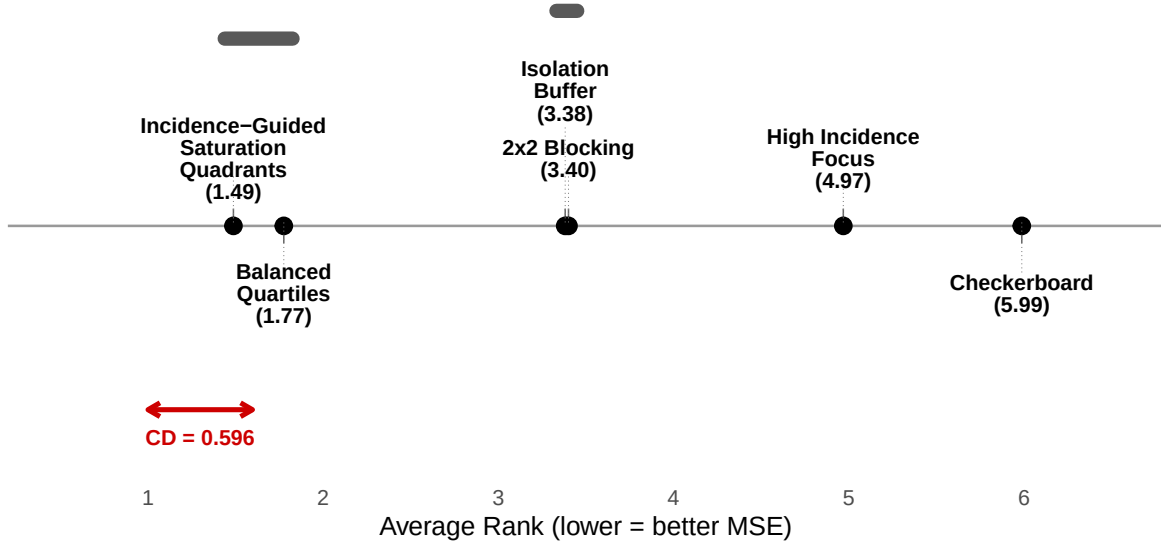
905 This section explains three results reported in the body of this chapter: why the bias of
906 Incidence-Guided Saturation Quadrants grew with τ , why the coverage of Checkerboard under
907 rook contiguity rose with τ , and why Checkerboard's two spillover regimes gave different MSEs
908 under rook contiguity. The stored results keep neither $\hat{\rho}$ nor the standard error of each fit, so we
909 re-ran every block for the six designs, both neighbor types, and all five values of τ , 2.4 million
910 fits of the oracle model, recording $\hat{\rho}$ and the standard error, and the re-run reproduced the

Critical Difference Diagram by Average Rank (6–design set, $\tau = 1.0$)



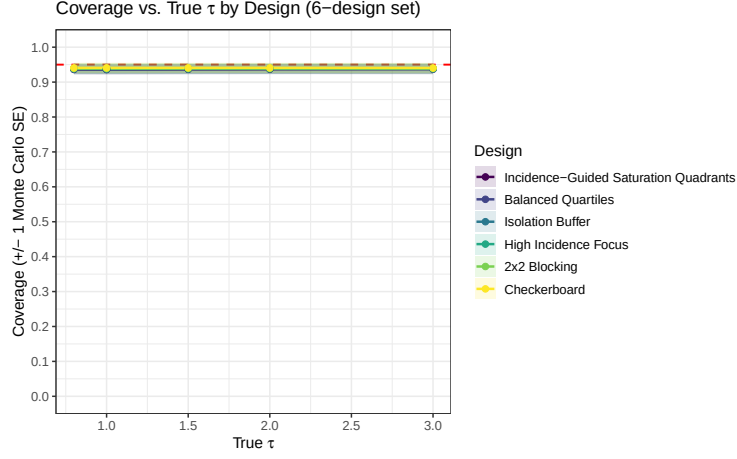
(a) Queen contiguity (primary)

Critical Difference Diagram by Average Rank (6–design set, $\tau = 1.0$)

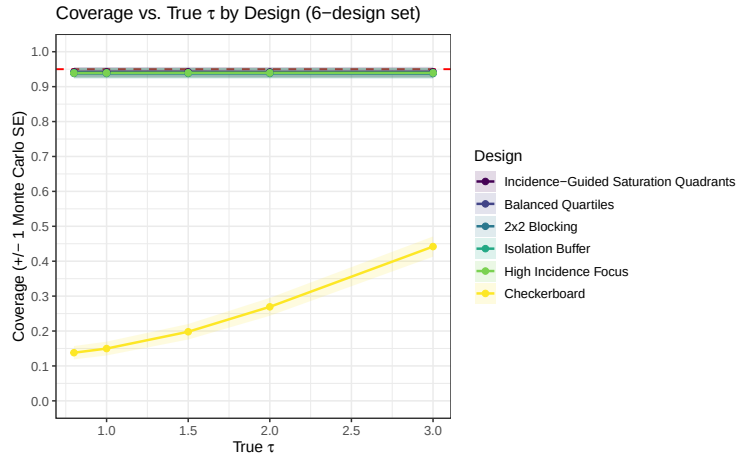


(b) Rook contiguity (sensitivity)

Figure 7: Nemenyi critical-difference diagrams for the six designs at $\tau = 1$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity), pooled over the five incidence configurations. Each design is placed at its average within-block MSE rank among the six designs over the 160 blocks of its neighbor type (1 = lowest MSE), given in parentheses. The red arrow shows the critical difference, 0.596 at $\alpha = 0.05$, and a gray bar joins designs whose pairwise Nemenyi comparisons were all not significant: in (a), Isolation Buffer with 2×2 Blocking, and 2×2 Blocking with Checkerboard and High Incidence Focus; in (b), Incidence-Guided Saturation Quadrants with Balanced Quartiles, and Isolation Buffer with 2×2 Blocking. Where two designs have nearly equal average ranks (Checkerboard and High Incidence Focus in (a), Isolation Buffer and 2×2 Blocking in (b)), the dotted leader line to the outer label crosses the neighboring design's label; the positions and labels are correct. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 8: Coverage of the 95% confidence intervals by design across the five values of the treatment effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each point averages the 160 scenarios of one design at one value of τ , pooled over the five incidence configurations. Shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of coverage; they describe the precision of a single scenario's coverage, not of the plotted average. The dashed red line marks the nominal 0.95 level. The legend lists the designs in order of pooled mean MSE at $\tau = 1$ for the neighbor type of the panel (Table 2). The vertical axis runs from 0 to 1, so lines close to 0.94 overlap: all six in (a), between 0.936 and 0.941, and all but Checkerboard in (b), between 0.937 and 0.942. The five values of τ share their assignments and noise. Under rook contiguity, τ is not identified for Checkerboard, the lowest line in (b), which the legend does not mark; its estimates are kept and flagged.

911 stored results in all 9,600 scenarios. Numbers below that involve $\hat{\rho}$, the standard error, or the
 912 terms of Equation 3 come from this re-run; the others come from the stored results. The third
 913 note also uses additional noise draws, described there.

914 **An exact decomposition of the error.** For one fit (k, j) , write Y , Z , and ϵ for Y_{kj} , Z_{kj} ,
 915 and ϵ_{kj} (Simulation design, resampling, and metrics). Let $V = [1, Z, S(Z), X_k]$ be the 100×4
 916 covariate matrix of the oracle model, so that Equation 1 reads $(I - \rho W)Y = V\theta + \epsilon$ with
 917 $\theta = (0, \tau, 1, \beta)^\top$. Let $A = (I - \rho W)^{-1}$ and $V^+ = (V^\top V)^{-1}V^\top$, and for a vector v write $[V^+v]_Z$
 918 for the coefficient on Z when v is regressed on the columns of V by least squares. Given $\hat{\rho}$, the
 919 estimated coefficients are $V^+(I - \hat{\rho}W)Y$ (Estimation). Because $(I - \hat{\rho}W)Y = V\theta + \epsilon + (\rho - \hat{\rho})WY$
 920 and $WY = WA(V\theta + \epsilon)$,

$$\hat{\tau} - \tau = [V^+\epsilon]_Z + (\rho - \hat{\rho})h + \tau(\rho - \hat{\rho})g, \quad (3)$$

921 where

$$g = [V^+WAZ]_Z, \quad h = [V^+WA\{S(Z) + \beta X_k + \epsilon\}]_Z.$$

922 The first term has mean zero given Z and X_k . The last term carries a factor of τ . It is
 923 present exactly when $g \neq 0$, where g is the coefficient on Z when the lagged treatment $WAZ =$
 924 $WZ + \rho W^2Z + \rho^2 W^3Z + \dots$ is regressed on the model's own covariates, and its sign and size are
 925 set by the error in $\hat{\rho}$. The concentrated likelihood depends on Y only through the residual sum
 926 of squares of $(I - \tilde{\rho}W)Y$ on V , and the part of that residual that involves τ is $\tau(\rho - \tilde{\rho})$ times the
 927 residual of WAZ on V . If WAZ lies in the column space of V , that part vanishes for every $\tilde{\rho}$,
 928 so $\hat{\rho}$ is the same at every value of τ ; if in addition $g = 0$, then $\hat{\tau} - \tau$ is the same at every value of
 929 τ . Equation 3 held on every re-run fit, with a largest residual of 4.2×10^{-13} ; for Checkerboard
 930 under rook contiguity it held in the reduced form given in the note on that design's coverage.

931 **Why the bias of Incidence-Guided Saturation Quadrants grew with τ .** Under queen

932 contiguity, pooled over the 160 blocks, the bias of Incidence-Guided Saturation Quadrants rose
 933 from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$ (Stability across treatment effect sizes), while that
 934 of Isolation Buffer stayed between 0.002 and 0.004. Almost all of the rise is the last term of
 935 Equation 3. In our fits, $\hat{\rho}$ was biased downward by about 0.06 on average (for Incidence-Guided
 936 Saturation Quadrants, by 0.071 at $\tau = 0.8$ and 0.054 at $\tau = 3$). We observed this downward
 937 bias in these simulations; we did not derive it or check it against a published finite-sample
 938 result. With $\rho - \hat{\rho}$ positive on average, the last term is positive and grows with τ for any
 939 design with $g > 0$. For Incidence-Guided Saturation Quadrants, g averaged 0.371 over the queen
 940 fits, and the last term contributed 0.021 of the bias at $\tau = 0.8$ and 0.043 at $\tau = 3$; the first
 941 term contributed 0.001 at both values, and the second 0.007 and 0.002. For Isolation Buffer, g
 942 averaged 0.020, and the last term contributed 0.001 and 0.004. Most of the difference in g comes
 943 from the control-only regime. Under the both-arms regime with $\rho = 0$, the lagged treatment is
 944 $WZ = S(Z)/\gamma$, one of the covariates, so $g = 0$ for every design. Under the control-only regime,
 945 $WZ = S(Z)/\gamma + Z \circ WZ$, and at $\rho = 0$, g is close to the average share of treated neighbors
 946 among treated clusters. That share is 0.57 for Incidence-Guided Saturation Quadrants, whose
 947 treated clusters are concentrated in the high-saturation quadrants, and exactly 0 for Isolation
 948 Buffer, whose treated clusters have no treated neighbors. For Isolation Buffer, $Z \circ WZ = 0$, so
 949 the two regimes define the same spillover covariate and the same g , which is 0 at $\rho = 0$ and
 950 about 0.06 at $\rho = 0.5$; at $\rho = 0$, its $\hat{\rho}$ and its estimation error were the same at every value of τ ,
 951 up to the optimizer's tolerance.

952 The same term explains why the control-only MSE of Incidence-Guided Saturation Quadrants
 953 rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$ while that of Isolation Buffer did not. Under that
 954 regime, averaged over the 80 queen blocks, the variance of the last term of Equation 3 rose from
 955 0.008 to 0.061 for Incidence-Guided Saturation Quadrants, while the variance of the rest of the
 956 error changed little (0.119 and 0.115). The variance of the last term grew more slowly than τ^2
 957 because $\hat{\rho}$ became less variable as τ grew. For Isolation Buffer, the variance of the last term

958 stayed below 0.0002. The rise in bias slowed at larger values of τ for a related reason: the part
 959 of the likelihood that involves τ sharpens it around the true ρ as τ grows, so the downward bias
 960 of $\hat{\rho}$ shrank.

961 **Why the coverage of Checkerboard under rook contiguity rose with τ .** Under rook
 962 contiguity, Checkerboard's coverage rose from 0.138 at $\tau = 0.8$ to 0.442 at $\tau = 3$ (Stability across
 963 treatment effect sizes), although its coefficient on Z estimates $\tau - \gamma$ at every value of τ . For this
 964 design $WZ = 1 - Z$, and because W is row-standardized, $W^2Z = W(1 - Z) = Z$. The powers of
 965 W applied to Z therefore alternate between $1 - Z$ and Z , and $WAZ = \{(1 - Z) + \rho Z\}/(1 - \rho^2)$
 966 lies in the span of the intercept and Z . The derivation of Equation 3 applies with the aliased
 967 spillover column dropped, $V = [1, Z, X_k]$ and $\theta = (\gamma, \tau - \gamma, \beta)^\top$, so that $\tau - \gamma$ takes the place of
 968 τ , and $g = -1/(1 + \rho)$. Two consequences follow: $\hat{\rho}$ is the same at every value of τ , and

$$\hat{\tau} - \tau = -\gamma + \delta + (\tau - \gamma) \frac{\hat{\rho} - \rho}{1 + \rho},$$

969 where δ collects the first two terms of that decomposition and, like $\hat{\rho}$, does not depend on τ . The
 970 error in $\hat{\rho}$ is therefore multiplied by $\tau - \gamma$, which grows with τ . The mean standard deviation
 971 of $\hat{\tau}$ rose from 0.202 at $\tau = 0.8$ to 0.327 at $\tau = 3$, and the standard error tracked it, rising on
 972 average from 0.199 to 0.322. With the bias held near $-\gamma$, wider intervals covered τ more often.
 973 The rise was largest at $\rho = 0$ (coverage 0.145 to 0.541) and smallest at $\rho = 0.5$ (0.130 to 0.265),
 974 consistent with the factor $1/(1 + \rho)$. The same term raised the MSE from 0.477 to 0.567. About
 975 three quarters of that increase was variance, $(n - 1)SD^2/n$ averaged over scenarios (Appendix
 976 A2), which rose from 0.041 to 0.109. The other quarter was squared bias, which rose from 0.436
 977 to 0.458 because $\hat{\rho}$ was biased slightly downward, moving the mean bias from -0.650 to -0.667 .
 978 The higher coverage at large τ reflects wider intervals around the wrong quantity, not a recovery
 979 of τ .

980 **Why Checkerboard's two spillover regimes gave different MSEs under rook contigu-**

ity. Under rook contiguity at $\tau = 1$, Checkerboard had an MSE of 0.477 under the both-arms regime and 0.483 under the control-only regime (Table 6), and a paired t -test of the difference over the 80 pairs of blocks, matched on incidence configuration, ρ , and γ , gives $p = 0.009$. For this design the two regimes define the same outcome model: because $WZ = 1 - Z$, both spillover covariates equal $\gamma(1 - Z)$ (Rook contiguity). The assignment is fixed and the incidence surfaces are shared, so the two blocks of a pair differ only in their noise, which is drawn from seed keys that include the regime (Table 12). The two MSEs therefore have the same expectation, and their difference is Monte Carlo error. The shared surfaces do not invalidate the paired test: given the surfaces, the two noise sets of a pair are exchangeable, so each pair's difference has conditional mean zero, and the 80 differences are uncorrelated. To check the test's calibration, we drew 400 further independent noise sets for these 80 pairs, with seed keys distinct from those of the simulation. At $\tau = 1$, the 80-block mean MSE had a standard deviation of 0.0020 across these noise sets, so the difference between two independent sets has a standard deviation of 0.0029; the observed difference, 0.0067, is 2.3 of these standard deviations. In paired t -tests between 200 disjoint pairs of the new noise sets, which by construction have no regime effect, p was at most 0.05 for 8% of the pairs and at most 0.009 for 2%. The test is therefore roughly calibrated, perhaps slightly liberal, and the observed difference is an unusually large Monte Carlo difference, of a size that arises about once in 50 such comparisons, not evidence of a regime effect. Most of it traces to the both-arms noise set, whose average of $(\hat{\rho} - \rho)/(1 + \rho)$ was higher than in 399 of the 400 new sets. The difference had the same sign at every value of τ and grew with τ . That adds no evidence: the five values of τ share their noise, and the error in $\hat{\rho}$ is multiplied by $\tau - \gamma$ (previous note). The argument extends to Isolation Buffer, whose two regimes also define the same outcome model (Sensitivity to spatial dependence, spillover magnitude, and spillover regime). Its assignments, unlike Checkerboard's, are redrawn for each regime, because their seed keys also include the regime (Table 12), but given the surfaces the assignments and noise of the two blocks of a pair are both exchangeable, so each pair's difference again has conditional mean

1007 zero. Its regime differences at $\tau = 1$ were not significant (paired t -test $p = 0.53$ under queen
1008 contiguity and $p = 0.12$ under rook contiguity).